

Fast Multi-Constellation GNSS Satellite Selection: Convex and Mixed-Integer Programming Approaches

Natnael S. Zewge[✉], Graduate Student Member, IEEE, and Hyochoong Bang[✉]

Abstract—A set of fast optimization-based procedures for the selection of an optimal subset of satellites from those within view of a receiver is provided. Resource limitations, the low marginal utility of using an all-in-view navigation solution, and the need for solution separation procedures in safety-critical navigation systems necessitate the use of a smaller subset from a collection of visible satellites. The goal of the present research is the fast selection of k satellites from among m that are available such that the selection is maximally informative (having minimum covariance measure) of user position. With many GNSS satellites currently deployed and more than four satellites required for integrity monitoring purposes, brute-force search through all ${}^m C_k$ options is not a reasonable approach. Satellite geometry and measurement precision are essential metrics in satellite selection; however, much of the literature on this topic has focused on geometry alone. In this work, we jointly account for satellite geometry and measurement precision in a single optimization framework. We present a semidefinite programming (SDP) formulation that considers both geometry and measurement quality. Afterward, we provide a reduction of the SDP form to that of a second-order cone program (SOCP). The SOCP formulation yields significant speed improvements while providing the same solution as the SDP form. Furthermore, we demonstrate exact satellite selection using a mixed-integer SOCP formulation, a first in the literature. Additionally, we show how context-specific constraints for GNSS can be easily incorporated into our selection framework. To illustrate the proposed set of methods, experiments are conducted on real-world multi-constellation GNSS data from GPS, GLONASS, BeiDou, Galileo, and QZSS satellites. For a comparative perspective, we offer an efficient greedy search technique, a genetic algorithm, a globally optimal exhaustive search (where applicable), and an all-in-view strategy.

Index Terms—Mixed-integer programming, multi-constellation GNSS, optimal experiment design, satellite selection, semidefinite programming (SDP), second-order cone programming (SOCP), weighted geometric dilution of precision (WGDP).

I. INTRODUCTION

A. Background

FULL operational capability of various GNSS constellations has allowed navigation and timing applications in numerous fields with high accuracy and availability [1], [2]. Nowadays, most receivers have at their disposal, more satellites than that

are required to compute a position solution [3]. Even when tracking a large number of satellites is possible with modern receivers, an all-in-view solution gives marginal benefits when compared to the performance of a carefully selected subset [3], [4], [5], [6], [7]. In some cases, an all-in-view solution may have degraded performance if the pseudo-range errors from some of the satellites are significant (due to faulty measurements for instance) [8, Ch. 17]. Therefore, an efficient selection algorithm that can eliminate measurement redundancy while incurring minimal performance penalty is desirable. Particular uses of satellite selection include Receiver Autonomous Integrity Monitoring (RAIM) [9], [10], [11], [12], [13], and in the transmission of correction and integrity parameters via VHF Data Broadcast (VDB) link in Ground-Based Augmentation Systems (GBAS) [14]. For solution separation RAIM in particular, the position solutions of multiple subsets of a given set of satellites are computed, and a chosen difference (for instance the maximum difference) in solution is compared against a threshold to indicate if a fault condition has occurred. Fault detection is an essential element for safety critical systems such as aircraft and satellite selection plays a critical role in such cases. With the emerging technology of low earth orbit positioning, navigation, and timing (LEO PNT) services which utilize a large number of satellites [15], and the growing set of GNSS constellations, the role of satellite selection will be important due to computational considerations [16].

The approaches taken by researchers to address the satellite selection problem range from early attempts that derive analytical solutions using geometry-based arguments to more sophisticated methods that use machine learning and evolutionary programming. The earliest accounts of geometry-based methods were stated in [17], [18]. The satellite geometric configuration that gives the largest volume of tetrahedron (hence the smallest geometric dilution of precision (GDOP)) was selected as the best arrangement. These methods were limited to four satellites. A selection method that utilized artificial neural networks was proposed in [19]. This paper introduced a technique wherein the authors estimate a functional form that maps eigenvalues that depend on the measurement matrix to GDOP. This functional form is learned from data, thereby, enabling the computation of GDOP without using matrix inversion. A computationally efficient recursive strategy for satellite selection was put forth in [20]. The key contribution in this work is a “revolving door” approach whereby one satellite is taken out of a selected subset while a new one is introduced. Subsequently, the matrix

Received 2 February 2024; revised 7 May 2024 and 20 November 2024; accepted 16 January 2025. Date of publication 27 January 2025; date of current version 20 June 2025. The review of this article was coordinated by Prof. Bin Lin. (Corresponding author: Hyochoong Bang.)

The authors are with the Department of Aerospace Engineering, Korea Advanced Institute Of Science and Technology, Daejeon 34141, South Korea (e-mail: nati_zewge@kaist.ac.kr; hcbang@kaist.ac.kr).

Digital Object Identifier 10.1109/TVT.2025.3534631

inversion lemma is used for GDOP computation which greatly reduces computational burden.

A greedy heuristic that works well in practice was explained in [3]. In this technique, satellites are removed one at a time, followed by an evaluation of corresponding geometries after removal. This iterative procedure stops when one reaches the required number of satellites, returning a solution that has a low GDOP. The authors in [21] presented a genetic algorithm where GDOP is the chosen objective to minimize. The work in [14] used an efficient heuristic for satellite selection for application in GBAS. The authors focused on keeping Vertical Protection Level (VPL) to a minimum as an objective, as is required in aviation applications. A rigorous approach to the satellite selection problem was presented in [22]. The authors demonstrated that the satellite subsets that form the largest polytope from a convex hull of endpoints of the vector from user to satellite, give the smallest GDOP. A data-driven method was introduced in [23]. The authors noted the importance of considering satellite geometry as well as pseudo-range measurement accuracy when selecting satellites. A score function approach was used to optimize for GDOP while logistic regression was used to predict large measurement errors. Other methods such as particle swarm optimization [24], Modified Beetle Antenna Search (MBAS) [25], and k -means clustering [26] have also been proposed.

B. Objective

The work presented herein is focused on providing a fast and flexible framework for the satellite selection problem. It is an extension of an earlier effort on satellite selection by the current authors [27], which was based on the work in [28]. The research in [28] gives a detailed account of the general source selection problem by formulating it as a convex optimization problem. Other authors have applied this technique for applications such as target tracking in wireless sensor networks [29], optimal sensor placement for state and parameter estimation for mechanical systems [30], and magnetic tracking systems [31]. The work in [28] is an extension of a related but different problem in statistics known as optimal experiment design [32], [33].

C. Contributions

1) A direct use of the tool in [28] will only allow for equal variance of measurements from sources (i.e. it only optimizes for source geometry) and derives a semidefinite programming (SDP) formulation. We extend this by assuming unequal variance of sources and derive detailed models for the variance of each source in a GNSS context (Section II-C). This results in a modified optimization objective more representative of real-world data acquisition. With this objective, we optimize for both source geometry and measurement precision. This extension has very important practical implications for GNSS, as satellites provide measurements with different variances. Assuming equal variances for GNSS satellites may lead to good geometry, but does not guarantee good performance in terms of carrier-to-noise ratio, and elevation-dependent atmospheric effects. Hence, we

work with a weighted optimization metric that balances several metrics in a single framework.

2) We provide a second-order cone programming (SOCP)–based reduction to the SDP-based method. This allows us to obtain solutions faster than the SDP method by more than an order of magnitude while getting the same solution. The method also exhibits several-fold speed improvements over a fast heuristic called greedy search and state-of-the-art evolutionary approaches. Furthermore, SOCPs can be implemented onboard embedded devices, which makes them very attractive for practical applications. In Section IV-B, we show comparative results of our reduction with that of SDP as well as greedy search and genetic algorithm. We further demonstrate our SOCP results on real-world data on an embedded platform.

3) We present an *exact* solution to the selection problem using a mixed-integer SOCP technique. This has been possible because of the proposed transformation from an SDP formulation into an SOCP (Ref. Section III-D2). The implications of this are significant because this approach definitively solves the selection problem. Furthermore, it can be used as a benchmarking tool for other suboptimal algorithms when brute-force global search is infeasible. This is a unique contribution of our work to the literature.

4) We provide a method to trade off vertical position error margins (for which we have strict requirements) and horizontal position errors (for which the margins are more relaxed). Our method gives a min-max optimization approach to give a reduced vertical positioning error. This result is useful for aviation applications where meeting vertical error requirements is of high importance for landing and approach.

The optimization approaches pursued in this work have the advantages of generality, speed, and the ability to incorporate constraints with ease. We note that the methods used in this research are useful not just in satellite selection but also in other measurement settings where multiple sources are considered. Measurements with the linear form, $\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{v}$, $\mathbf{v} \sim \mathcal{N}(\mathbf{0}, \mathbf{R})$ (or a linearized counterpart of a nonlinear observation) are candidates for the application of our fast and/or exact optimization frameworks. Accordingly, applications such as wireless sensor networks, optimal sensor placement for environmental and industrial machinery monitoring, UAV coordination and swarming for maximally informative coverage, and optimal feature selection in feature-based navigation in mobile robotic systems are some of the major application areas in which the tools of this work can be used.

D. Paper Outline

The rest of the paper is organized as follows: Section II presents the satellite pseudo-range measurement equations and their linearization leading to the canonical linear estimation problem. Detailed models of errors affecting pseudo-range measurements are derived. Further in this section, a succinct summary of the research problem is presented. In Section III, a short introduction to semidefinite programming and second-order cone programming is presented, followed by a formulation of the satellite selection problem. A lemma that enables the

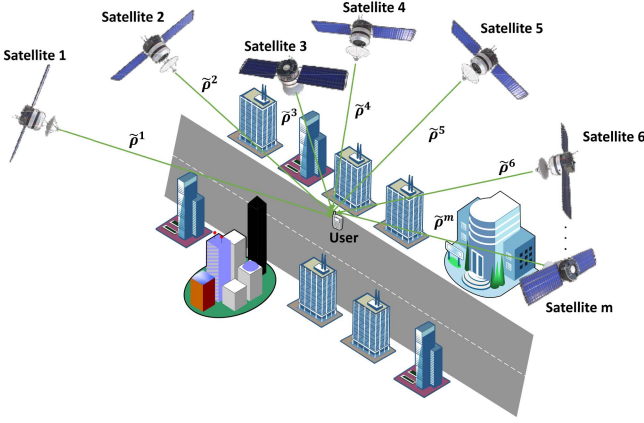


Fig. 1. Typical urban GNSS measurement scenario.

conversion of the SDP form of the selection problem to an SOCP is presented in this section. Section IV presents our experimental setup, results, and discussion. In this section, we also show how the standard optimization formulation can be modified to address additional use cases. Concluding remarks are presented in Section V. The appendices contain additional information that supplements the main text.

II. PROBLEM FORMULATION

A. GNSS Navigation Equations

A set of m navigation satellites provide pseudo-range measurements to the user antenna phase center according to the following set of nonlinear relations [34]

$$\tilde{\rho}^i = \rho^i + c\delta t_u - c\delta t^i + I_u^i + T_u^i + \epsilon^i, \quad i = 1, \dots, m, \quad (1)$$

where the subscript i indicates the i -th satellite. The speed of light is indicated by c , δt_u is the user clock offset, δt^i is the satellite clock offset, and I_u^i and T_u^i indicate delays due to ionosphere and troposphere respectively. The term ϵ is an aggregate of errors due to multipath, non-line-of-sight (NLOS), and receiver noise. A basic measurement scenario is illustrated in Fig. 1.

Correcting for delays caused by ionospheric and tropospheric delays, as well as satellite clock offset yields the form

$$\tilde{\rho}_c^i = \rho^i + c\delta t_u + \bar{\epsilon}^i \quad i = 1, \dots, m, \quad (2)$$

where $\tilde{\rho}_c^i$ is the corrected pseudo-range and $\bar{\epsilon}^i$ models all the residual errors after correction. ρ^i is the true distance from user to satellite as measured by the ℓ_2 -norm

$$\rho^i = \|\mathbf{x}^i - \mathbf{x}\|_2, \quad i = 1, \dots, m. \quad (3)$$

The position (column)vectors $\mathbf{x}^i = (x^i, y^i, z^i)$ and $\mathbf{x} = (x, y, z)$, reported in the Earth Centered Earth Fixed (ECEF) frame, are satellite position and user position respectively.

A vector of estimated user position and clock bias is denoted by $\hat{\rho} = (\hat{\mathbf{x}}, c\hat{\delta}t_u)$. Therefore, the estimated pseudo-range is given by

$$\hat{\rho}^i = \|\mathbf{x}^i - \hat{\mathbf{x}}\|_2 + c\hat{\delta}t_u, \quad i = 1, \dots, m. \quad (4)$$

A first-order Taylor expansion of (2) about the current estimate gives the relation

$$\Delta \rho = \begin{bmatrix} \Delta \rho^1 \\ \Delta \rho^2 \\ \vdots \\ \Delta \rho^m \end{bmatrix} = \begin{bmatrix} (-\ell^{(1)})^T & 1 \\ (-\ell^{(2)})^T & 1 \\ \vdots & \vdots \\ (-\ell^{(m)})^T & 1 \end{bmatrix} \begin{bmatrix} \Delta \mathbf{x} \\ \Delta c\delta t_u \end{bmatrix} + \begin{bmatrix} \bar{\epsilon}^1 \\ \bar{\epsilon}^2 \\ \vdots \\ \bar{\epsilon}^m \end{bmatrix}, \quad (5)$$

where $\Delta \rho^i = \tilde{\rho}_c^i - \hat{\rho}^i$, $\Delta c\delta t_u = c\delta t_u - c\hat{\delta}t_u$, and $\Delta \mathbf{x} = \mathbf{x} - \hat{\mathbf{x}}$. The term $(\ell^{(i)})^T$ is the line-of-sight unit vector from the user's estimated location to the i -th satellite and is given by

$$\ell^{(i)} = \frac{\mathbf{x}^i - \hat{\mathbf{x}}_u}{\|\mathbf{x}^i - \hat{\mathbf{x}}_u\|}, \quad i = 1, \dots, m. \quad (6)$$

We define the matrix

$$\begin{bmatrix} (-\ell^{(1)})^T & 1 \\ (-\ell^{(2)})^T & 1 \\ \vdots & \vdots \\ (-\ell^{(m)})^T & 1 \end{bmatrix} = \mathbf{H} \in \mathbb{R}^{m \times n}, \quad (7)$$

as the geometry matrix in the ECEF frame (with $n = 4$).¹ We therefore have the pseudo-range equation

$$\Delta \rho = \mathbf{H} \Delta \mathbf{x}_a + \bar{\epsilon}, \quad (8)$$

where we define the augmented state $\Delta \mathbf{x}_a = (\Delta \mathbf{x}, \Delta c\delta t_u) \in \mathbb{R}^n$ and $\Delta \rho, \bar{\epsilon} \in \mathbb{R}^m$. The solution to (8) gets added to the known initial conditions to produce a new solution. We assume that the error vector $\bar{\epsilon}$ satisfies $\mathbb{E}[\bar{\epsilon}] = 0$ and its elements are uncorrelated (i.e. $\mathbb{E}[\bar{\epsilon}_i \bar{\epsilon}_j] = 0, \forall i \neq j$).²

B. Solution to the GNSS Navigation Equations

For $\bar{\epsilon} = (\bar{\epsilon}^1, \bar{\epsilon}^2, \dots, \bar{\epsilon}^m)$, the measurement error covariance matrix, $\mathbf{R} \in \mathbb{S}_{++}^m$, is given by $\mathbb{E}[\bar{\epsilon} \bar{\epsilon}^T]$, where $\mathbb{S}_{++}^m$ is the set of symmetric positive definite matrices. With $\mathbf{R}^{-1}$ as the weight matrix, the weighted least squares approximate solution to (8) is given by

$$\Delta \hat{\mathbf{x}}_a = (\mathbf{H}^T \mathbf{R}^{-1} \mathbf{H})^{-1} \mathbf{H}^T \mathbf{R}^{-1} \Delta \rho. \quad (9)$$

The estimation error δ is given by

$$\delta = \Delta \hat{\mathbf{x}}_a - \Delta \mathbf{x}_a = (\mathbf{H}^T \mathbf{R}^{-1} \mathbf{H})^{-1} \mathbf{H}^T \mathbf{R}^{-1} \bar{\epsilon}. \quad (10)$$

The estimation error covariance is given by

$$\mathbf{P} = \mathbb{E}[\delta \delta^T], \quad (11)$$

for which we have the form (this is a standard result in linear estimation [36])

$$\mathbf{P} = (\mathbf{H}^T \mathbf{R}^{-1} \mathbf{H})^{-1}, \quad (12)$$

¹In applications, we usually use vertical and horizontal errors to describe user position uncertainty, necessitating a local tangent frame description [34], [35]. In this case, the line of sight vector to the i -th satellite is $(-\cos \theta^i \cos \psi^i, -\sin \theta^i \cos \psi^i, \sin \psi^i)$, with θ^i and ψ^i representing azimuth and elevation to the i -th satellite respectively. We use this form in our experiments.

²Such assumptions on the error vector do not exactly hold in reality but are used as a standard simplifying assumption in the literature due to the difficulty of modeling the correlations across satellites [34]

which gives a measure of the accuracy of our estimates. $\mathbf{H}$ is the function of satellite geometry and $\mathbf{R}$ is a measure of measurement precision. The matrix $\mathbf{R}^{-1}$ has the form

$$\mathbf{R}^{-1} = \begin{bmatrix} \frac{1}{\sigma_{\bar{\epsilon}^1}^2} & & & \\ & \frac{1}{\sigma_{\bar{\epsilon}^2}^2} & & \\ & & \ddots & \\ & & & \frac{1}{\sigma_{\bar{\epsilon}^m}^2} \end{bmatrix} \in S_{++}^m, \quad (13)$$

where $\sigma_{\bar{\epsilon}^i}^2$ is the error variance of the measurement from the i -th satellite. The corresponding reciprocal is a measure of the information contained in the i -th satellite's measurement. A normalized version of this matrix is used in our final analysis to obtain a dimensionless optimization objective.

C. Errors Affecting Pseudo-Range Measurements

The value of the variance $\sigma_{\bar{\epsilon}^i}^2$ of each satellite measurement depends on several factors [37, Ch. 11], [8, Ch. 9], [38]. The signal from each satellite experiences varying degrees of influence due to ionospheric and tropospheric refraction, multipath, and receiver noise. In addition, the user range accuracy (URA) factors into the calculation of error contributions. We therefore have, for each satellite,

$$\sigma_{\bar{\epsilon}^i}^2 = \sigma_{\bar{\epsilon}^i,URA}^2 + \sigma_{\bar{\epsilon}^i,iono}^2 + \sigma_{\bar{\epsilon}^i,tropo}^2 + \sigma_{\bar{\epsilon}^i,mp}^2 + \sigma_{\bar{\epsilon}^i,rcvn}^2, \quad (14)$$

where $\sigma_{\bar{\epsilon}^i,URA}^2$ is the variance of the user range error, $\sigma_{\bar{\epsilon}^i,rcvn}^2$ is the error variance contribution from the receiver noise, $\sigma_{\bar{\epsilon}^i,iono}^2$, $\sigma_{\bar{\epsilon}^i,tropo}^2$, and $\sigma_{\bar{\epsilon}^i,mp}^2$ are the variances due to the effects of ionosphere, troposphere, and multipath respectively. Details on the aforementioned variances are discussed below.

1) *User Range Accuracy (URA)*: URA represents 1-sigma range errors to the satellite caused due to satellite and control segment errors. This value is transmitted as part of the satellite navigation message. The values for this quantity commonly range from 1 – 2 meters.

2) *Ionospheric Propagation Error*: GNSS signals experience refraction when propagating through the atmosphere. Upon observing Fig. 2, one can see that this effect is more pronounced for lower-elevation satellites because the signals pass through a larger portion of the atmosphere. The ionospheric propagation delay is related to the satellite elevation angle ψ . Assuming a single-frequency receiver and ignoring higher-order frequency effects, the ionospheric delay estimation error is given by

$$\sigma_{\bar{\epsilon}^i,iono} = \left[1 - \left(\frac{R \cos(\psi^i)}{R + h_I} \right)^2 \right]^{-\frac{1}{2}} \sigma_v, \quad (15)$$

where

$$\sigma_v = \begin{cases} 9m, & 0 \leq |\phi| \leq 20^\circ \\ 4.5m, & 20^\circ < |\phi| \leq 55^\circ \\ 6m, & 55^\circ > |\phi|, \end{cases} \quad (16)$$

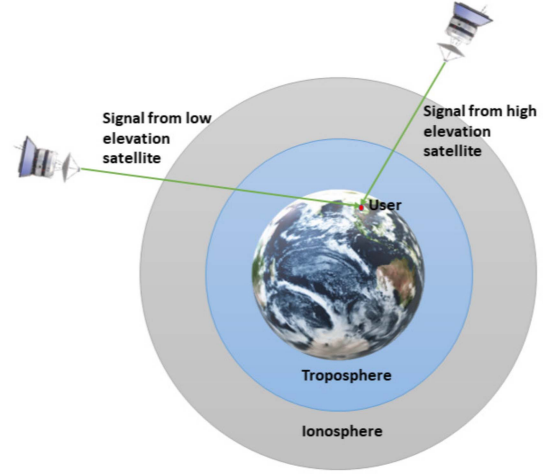


Fig. 2. Signal propagation to the user in ionosphere and troposphere for low- and high-elevation satellites.

with ϕ representing the user latitude. R is the mean radius of the Earth which is 6378.14 km, and h_I is the mean ionospheric height which is taken to be 350 km.

3) *Tropospheric Propagation Error*: A simplified tropospheric delay estimation error model is given by

$$\sigma_{\bar{\epsilon}^i,tropo} = \frac{1.001}{\sqrt{0.002001 + \sin^2(\psi^i)}} \sigma_t, \quad (17)$$

where $\sigma_t = 0.12m$. The first term in the product on the right-hand side of (17) is the Black and Eisner mapping. Fig. 2 shows the scenario where signals from low-elevation satellites pass through a bigger portion of the troposphere, hence leading to a larger error. The model is simplified in that it does not take into account effects such as temperature, pressure, and day of the year.

4) *Multipath*: GNSS satellite elevation has a direct relation with multipath. Low-elevation satellite signals are more likely to be received through reflections from the ground or structures surrounding the user than signals coming from high elevations. This relationship between multipath error and elevation ψ in degrees is represented by

$$\sigma_{\bar{\epsilon}^i,mp} = 0.13 + 0.53 \exp\left(-\frac{1}{10}\psi^i\right). \quad (18)$$

A typical multipath scenario is shown in Fig. 3. We note that a variety of other factors influence multipath error, such as the location of the receiver and the type of antenna. Here we focus only on the elevation dependence of the error.

5) *Tracking Errors Due to Receiver Noise*: Code tracking errors within receivers are calculated by a discriminator function. This function can have errors due to thermal noise, external RF interference, and other GNSS signals transmitting on the same frequency. Furthermore, this function is dependent upon the carrier-to-noise ratio (C/N_0). For an early-minus-late power discriminator, the code tracking noise standard deviation in

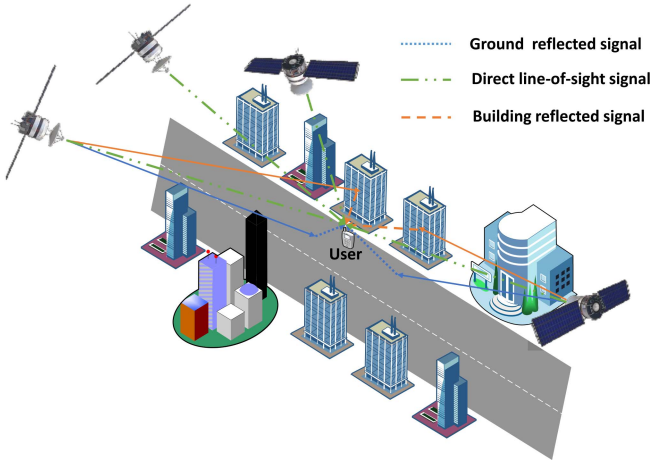


Fig. 3. Multipath interference due to reflections from the ground and urban structures.

chips, σ_{rc} , is given by

$$\sigma_{rc} = \sqrt{\frac{B_{LCO}d}{2(C/N_0)} \left[1 + \frac{2}{(2-d)(C/N_0)(\tau_a)} \right]}, \quad (19)$$

where B_{LCO} is the double-sided noise bandwidth of the code tracking loop, and d is the correlator spacing. τ_a is the correlator accumulation interval. The pseudo-range error standard deviation in meters due to tracking noise is given by

$$\sigma_{\hat{e}^i,rcvn} = \frac{c}{f_{co}} \sigma_{rc}, \quad (20)$$

where c is the speed of light as previously defined and f_{co} is the spreading-code chipping rate which is expressed as a multiple of 1.023×10^6 chips/s. We take $B_{LCO} = 2$ Hz, $d = 0.1$ chips, and $\tau_a = 0.02$ sec. for our case.

We have now accounted for the errors indicated in (14). Subsequent sections will show how these factors combined with satellite geometry, are incorporated into an optimization-based satellite selection procedure.

D. Problem Statement

Given m satellites in view of a receiver, the goal in this research is to select k satellites, where $4 \leq k < m$, such that some measure of the estimation error covariance given in (12), is minimized. We see that the estimation error covariance depends on the geometry matrix $\mathbf{H}$ and the measurement error covariance matrix $\mathbf{R}$. A standard metric used in the literature for satellite selection is to minimize the GDOP which is defined as

$$\text{GDOP} = \sqrt{\text{Tr}(\mathbf{H}^T \mathbf{H})^{-1}}. \quad (21)$$

A modified metric that accounts for the variation in measurement error variance [39] (discussed in II-C) and satellite geometry is

called the Weighted GDOP (WGDOP) and has the form³

$$\text{WGDOP} = \sqrt{\text{Tr}(\mathbf{H}^T \mathbf{R}^{-1} \mathbf{H})^{-1}}. \quad (22)$$

The k satellites that minimize (22), form the optimal subset.

III. METHODOLOGY

A. Overview of Semidefinite Programming

The set of symmetric $n \times n$ matrices is denoted by

$$\mathbb{S}^n = \{X \in \mathbb{R}^{n \times n} \mid X = X^T\}. \quad (23)$$

A standard inequality form semidefinite program is a convex optimization problem which is written as

$$\begin{aligned} & \underset{\mathbf{x} \in \mathbb{R}^n}{\text{minimize}} \quad \mathbf{c}^T \mathbf{x} \\ & \text{subject to} \quad \mathbf{F}_0 + \sum_{i=1}^n x_i \mathbf{F}_i \succeq 0, \quad i = 1, \dots, n, \end{aligned} \quad (24)$$

where $\succeq$ denotes a generalized inequality and where $\mathbf{F}_i \in \mathbb{S}^m$ and $\mathbf{c} \in \mathbb{R}^n$.

B. Overview of Second-Order Cone Programming

A second-order cone program (SOCP) is a convex optimization problem with the form

$$\begin{aligned} & \underset{\mathbf{x} \in \mathbb{R}^n}{\text{minimize}} \quad \mathbf{f}^T \mathbf{x} \\ & \text{subject to} \quad \|\mathbf{A}_i \mathbf{x} + \mathbf{b}_i\|_2 \leq \mathbf{c}_i^T \mathbf{x} + d_i, \quad i = 1, \dots, p, \\ & \quad \mathbf{F} \mathbf{x} = \mathbf{g}, \end{aligned} \quad (25)$$

where $\mathbf{A}_i \in \mathbb{R}^{n_i \times n}$, $\mathbf{F} \in \mathbb{R}^{q \times n}$, $\mathbf{b}_i \in \mathbb{R}^{n_i}$, $\mathbf{c}_i \in \mathbb{R}^n$, and $d_i \in \mathbb{R}$.

There exist efficient interior-point methods that can solve problems of the form (24) and (25) to find a global solution.

C. Scalar Metrics Associated With Covariance Matrices

Consider the covariance matrix $\mathbf{P}$ stated in (12). Taking the commonly used Gaussian model, the probability density function of the estimation error, δ , is given by

$$f(\delta) = \frac{1}{(2\pi)^{\frac{n}{2}}} \det(\mathbf{P})^{-\frac{1}{2}} \exp\left(-\frac{1}{2} \delta^T \mathbf{P}^{-1} \delta\right). \quad (26)$$

We state two scalar metrics returned from functions of the form $f: \mathbb{S}^n \rightarrow \mathbb{R}$, with $\text{dom } f = \mathbb{S}_+^n$, and with the form

$$f(\mathbf{P}) = f(\mathbf{H}^T \mathbf{R}^{-1} \mathbf{H})^{-1}. \quad (27)$$

1) *Metric 1:* The first metric is the trace of the error covariance matrix which signifies the mean squared estimation error. It is given by

$$\mathbb{E}[\delta^T \delta] = \mathbb{E}[\text{Tr}(\delta \delta^T)] = \text{Tr}(\mathbf{P}) = \text{Tr}(\mathbf{H}^T \mathbf{R}^{-1} \mathbf{H})^{-1}. \quad (28)$$

we note here the similarity with the WGDOP metric in (22). This metric is termed as A -optimality criterion.

³We normalize $\mathbf{R}^{-1}$ with the norm of the vector of its diagonal elements. This makes the WGDOP metric dimensionless while still maintaining the relative importance of the measurement precision values.

2) *Metric 2*: This metric is returned from the maximum eigenvalue function of the error covariance matrix, which is a measure of the worst-case error variance. It takes the form

$$\lambda_{max}(\mathbf{P}) = \lambda_{max}(\mathbf{H}^T \mathbf{R}^{-1} \mathbf{H})^{-1}, \quad (29)$$

where λ_{max} is the maximum eigenvalue function. This has the term *E*-optimality criterion. The two metrics will be put into context in Sections III-D and IV-C respectively.

D. Satellite Selection as an Optimization Problem

In the first half of this subsection, we present an SDP approach that jointly takes into account the geometry and measurement accuracy of satellites. The error models presented in Section II-C have an impact on which subsets are selected within the optimization framework. In the second half, we derive an SOCP approach that gives significant computational benefits compared to the SDP method. Lastly, an exact solution to the selection problem is obtained via a mixed-integer SOCP formulation.

1) *SDP Formulation With Weighted Objective*: Let the i -th row of the measurement matrix $\mathbf{H}$ be $\mathbf{h}_i^T \in \mathbb{R}^n$, such that

$$\mathbf{H} = \begin{bmatrix} - & \mathbf{h}_1^T & - \\ - & \mathbf{h}_2^T & - \\ & \vdots & \\ - & \mathbf{h}_m^T & - \end{bmatrix}. \quad (30)$$

Then, the covariance $\mathbf{P}$ becomes

$$\mathbf{P} = \left(\sum_{i=1}^m \frac{1}{\sigma_{\bar{\epsilon}^i}^2} \mathbf{h}_i \mathbf{h}_i^T \right)^{-1}. \quad (31)$$

The desired satellite selection can be posed as a Boolean optimization problem with the form

$$\begin{aligned} & \underset{\mu_1, \mu_2, \dots, \mu_m}{\text{minimize}} && f \left(\sum_{i=1}^m \mu_i \frac{1}{\sigma_{\bar{\epsilon}^i}^2} \mathbf{h}_i \mathbf{h}_i^T \right)^{-1} \\ & \text{subject to} && \sum_{i=1}^m \mu_i = k, \\ & && \mu_i \in \{0, 1\}, \quad i = 1, \dots, m, \end{aligned} \quad (32)$$

with $\boldsymbol{\mu} = (\mu_1, \dots, \mu_m)$. $\mu_i = 1$ indicates inclusion of the i -th satellite in the selection, whereas $\mu_i = 0$ represents exclusion of the i -th satellite from the selection. Our formulation extends the work in [28] by including the weight factor $1/\sigma_{\bar{\epsilon}^i}^2$, derived in Section II-C. This allows satellites to be selected not only based on geometry but also on measurement precision. *Metric 1* in Section III-C parallels the WGDOP objective we wish to optimize. We therefore pursue this optimization objective for our immediate discussion. Section IV-C discusses the benefits of using *Metric 2* for satellite selection in safety-critical systems. Using the metric in (28), we have *A*-optimal selection problem

with weighted objective given by

$$\begin{aligned} & \underset{\mu_1, \mu_2, \dots, \mu_m}{\text{minimize}} && \text{Tr} \left(\sum_{i=1}^m \mu_i \frac{1}{\sigma_{\bar{\epsilon}^i}^2} \mathbf{h}_i \mathbf{h}_i^T \right)^{-1} \\ & \text{subject to} && \sum_{i=1}^m \mu_i = k, \\ & && \mu_i \in \{0, 1\}, \quad i = 1, \dots, m. \end{aligned} \quad (33)$$

This form involves the constraint $\mu_i \in \{0, 1\}$, which is highly nonconvex. Applying a relaxation procedure which converts $\mu_i \in \{0, 1\}$ to $0 \leq \mu_i \leq 1$ yields the problem

$$\begin{aligned} & \underset{\mu_1, \mu_2, \dots, \mu_m}{\text{minimize}} && \text{Tr} \left(\sum_{i=1}^m \mu_i \frac{1}{\sigma_{\bar{\epsilon}^i}^2} \mathbf{h}_i \mathbf{h}_i^T \right)^{-1} \\ & \text{subject to} && \sum_{i=1}^m \mu_i = k, \\ & && 0 \leq \mu_i \leq 1, \quad i = 1, \dots, m. \end{aligned} \quad (34)$$

This is a convex optimization problem and hence tractable. The relaxation procedure brings with it a degree of suboptimality. However, it has been shown elsewhere that the suboptimality gap for such relaxations is usually very small [40]. Moreover, the authors in [28] show that for large m the relaxation gives very good results. The solution to (34) gives an approximate maximally informative set of k satellites considering GDOP and measurement accuracy simultaneously. It is often desirable to state (34) in an epigraph form. Introducing a variable $\mathbf{u} \in \mathbb{R}^n$

$$\begin{aligned} & \underset{\substack{\mu_1, \mu_2, \dots, \mu_m \\ u_1, u_2, \dots, u_n}}{\text{minimize}} && \sum_{j=1}^n u_j \\ & \text{subject to} && \begin{bmatrix} \sum_{i=1}^m \mu_i \frac{1}{\sigma_{\bar{\epsilon}^i}^2} \mathbf{h}_i \mathbf{h}_i^T & \mathbf{e}_j \\ \mathbf{e}_j^T & u_j \end{bmatrix} \succeq 0, \quad j = 1, \dots, n, \\ & && \sum_{i=1}^m \mu_i = k, \\ & && 0 \leq \mu_i \leq 1, \quad i = 1, \dots, m, \end{aligned} \quad (35)$$

where $\mathbf{e}_j$ is the j -th unit vector. This is an SDP, the derivation of which is stated in Appendix A. Standard convex optimization solvers can be used to obtain its solution. The solution is a vector $\boldsymbol{\mu}^* \in \mathbb{R}^m$. Each μ_i^* , where $\mu_i^* \in \boldsymbol{\mu}^*$ is a number in $[0, 1]$. Selecting the k largest values of $\boldsymbol{\mu}^*$ gives the k best satellites considering measurement quality and satellite geometry concurrently.

2) *SOCP Formulation With Weighted Objective*: The authors credit the work in [32], [41], [42] for background information on experiment design problems and the derivation of tractable optimization approaches for them; however, the approach for the SOCP formulation for the approximate and exact selection problem (i.e. using Lemma 1) is our own.⁴ Consider the problem

⁴We note that, although related, the experimental design problem and the selection problem are different.

in (34). An equivalent problem can be stated as

$$\begin{aligned} & \underset{\mu_1, \mu_2, \dots, \mu_m}{\text{minimize}} \quad \sum_{j=1}^n e_j^T \left(\sum_{i=1}^m \mu_i \frac{1}{\sigma_{\bar{\epsilon}^i}^2} \mathbf{h}_i \mathbf{h}_i^T \right)^{-1} e_j \\ & \text{subject to} \quad \sum_{i=1}^m \mu_i = k, \\ & \quad 0 \leq \mu_i \leq 1, \quad i = 1, \dots, m. \end{aligned} \quad (36)$$

This follows from the property

$$\text{Tr} \left(\sum_{i=1}^m \mu_i \frac{1}{\sigma_{\bar{\epsilon}^i}^2} \mathbf{h}_i \mathbf{h}_i^T \right)^{-1} = \sum_{j=1}^n e_j^T \left(\sum_{i=1}^m \mu_i \frac{1}{\sigma_{\bar{\epsilon}^i}^2} \mathbf{h}_i \mathbf{h}_i^T \right)^{-1} e_j. \quad (37)$$

Setting

$$\mathbf{Q} = \left(\sum_{i=1}^m \mu_i \frac{1}{\sigma_{\bar{\epsilon}^i}^2} \mathbf{h}_i \mathbf{h}_i^T \right) = \mathbf{H}^T \text{diag}(\boldsymbol{\mu}) \mathbf{R}^{-1} \mathbf{H}, \quad (38)$$

with $\mathbf{Q} \in \mathbb{S}^n$ and $\boldsymbol{\mu} = (\mu_1, \dots, \mu_m)$, taking the square root of $\mathbf{R}^{-1}$ as $\mathbf{R}^{-\frac{1}{2}}$, and defining $\bar{\mathbf{H}} = \mathbf{R}^{-\frac{1}{2}} \mathbf{H}$ we have⁵

$$\begin{aligned} & \underset{\mu_1, \mu_2, \dots, \mu_m}{\text{minimize}} \quad \sum_{j=1}^n e_j^T \mathbf{Q}^{-1} e_j \\ & \text{subject to} \quad \mathbf{Q} = \bar{\mathbf{H}}^T \text{diag}(\boldsymbol{\mu}) \bar{\mathbf{H}} \\ & \quad \sum_{i=1}^m \mu_i = k, \\ & \quad 0 \leq \mu_i \leq 1, \quad i = 1, \dots, m. \end{aligned} \quad (39)$$

To facilitate the conversion to an SOCP, we eliminate the summation in the objective of (39) by stacking n unit vectors $\mathbf{e}$ to form the vector $\tilde{\mathbf{e}} = (e_1, \dots, e_n) \in \mathbb{R}^{n \times 1}$ from which $\tilde{\mathbf{e}}^T = (e_1, \dots, e_n)^T \in \mathbb{R}^{1 \times n}$. With $r = mn$, we now state

Problem 1:

$$\begin{aligned} & \underset{\tilde{\mu}_1, \tilde{\mu}_2, \dots, \tilde{\mu}_r}{\text{minimize}} \quad \tilde{\mathbf{e}}^T \tilde{\mathbf{Q}}^{-1} \tilde{\mathbf{e}} \\ & \text{subject to} \quad \tilde{\mathbf{Q}} = \tilde{\mathbf{H}}^T \text{diag}(\tilde{\boldsymbol{\mu}}) \tilde{\mathbf{H}} \\ & \quad \sum_{i=1}^r \tilde{\mu}_i = nk, \\ & \quad 0 \leq \tilde{\mu}_i \leq 1, \quad i = 1, \dots, r, \end{aligned} \quad (40)$$

with $\tilde{\boldsymbol{\mu}} = (\tilde{\mu}_1, \dots, \tilde{\mu}_r)$ and where we define the block diagonal matrices $\tilde{\mathbf{Q}} = \text{blkdiag}(\mathbf{Q}, \dots, \mathbf{Q}) \in \mathbb{S}^{n^2}$ and $\tilde{\mathbf{H}} = \text{blkdiag}(\bar{\mathbf{H}}, \dots, \bar{\mathbf{H}}) \in \mathbb{R}^{r \times n^2}$. We can now proceed to the major steps toward conversion to an SOCP. With the introduction of a vector of variables $\mathbf{z} = (z_1, \dots, z_m)$ and $\tilde{\mathbf{z}} = (\mathbf{z}, \dots, \mathbf{z}) \in \mathbb{R}^r$, let us define the following problem

Problem 2:

$$\begin{aligned} & \underset{\substack{\tilde{\mu}_1, \tilde{\mu}_2, \dots, \tilde{\mu}_r \\ \tilde{z}_1, \tilde{z}_2, \dots, \tilde{z}_r}}{\text{minimize}} \quad \sum_{i=1}^r \frac{\tilde{z}_i^2}{2\tilde{\mu}_i} \\ & \text{subject to} \quad \tilde{\mathbf{H}}^T \tilde{\mathbf{z}} = \tilde{\mathbf{e}}, \\ & \quad \sum_{i=1}^r \tilde{\mu}_i = nk, \\ & \quad 0 \leq \tilde{\mu}_i \leq 1, \quad i = 1, \dots, r. \end{aligned} \quad (41)$$

Lemma 1: Problems 1 and 2 are equivalent.

Proof: Let $\mathbf{W} = \text{diag}(\tilde{\mu}_1, \dots, \tilde{\mu}_r) \in \mathbb{S}_+^r$. Problem 2 is an equality-constrained quadratic program that can be stated in the form

$$\begin{aligned} & \underset{\substack{\tilde{\mu}_1, \tilde{\mu}_2, \dots, \tilde{\mu}_r \\ \tilde{\mathbf{z}} \in \mathbb{R}^r}}{\text{minimize}} \quad \frac{1}{2} \tilde{\mathbf{z}}^T \mathbf{W}^{-1} \tilde{\mathbf{z}} \\ & \text{subject to} \quad \tilde{\mathbf{H}}^T \tilde{\mathbf{z}} = \tilde{\mathbf{e}}, \\ & \quad \sum_{i=1}^r \tilde{\mu}_i = nk, \\ & \quad 0 \leq \tilde{\mu}_i \leq 1, \quad i = 1, \dots, r. \end{aligned} \quad (42)$$

We hold $\tilde{\boldsymbol{\mu}}$ constant and optimize over $\tilde{\mathbf{z}}$. The Lagrangian of (42) is

$$L(\tilde{\mathbf{z}}, \boldsymbol{\lambda}) = \frac{1}{2} \tilde{\mathbf{z}}^T \mathbf{W}^{-1} \tilde{\mathbf{z}} + \boldsymbol{\lambda}^T (\tilde{\mathbf{H}}^T \tilde{\mathbf{z}} - \tilde{\mathbf{e}}), \quad (43)$$

where $\boldsymbol{\lambda}$ is the Lagrange multiplier. Taking the gradient of (43), one finds

$$\nabla L(\tilde{\mathbf{z}}, \boldsymbol{\lambda}) = \begin{bmatrix} \mathbf{W}^{-1} & \tilde{\mathbf{H}} \\ \tilde{\mathbf{H}}^T & \mathbf{0} \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{z}} \\ \boldsymbol{\lambda} \end{bmatrix} + \begin{bmatrix} \mathbf{0} \\ -\tilde{\mathbf{e}} \end{bmatrix}. \quad (44)$$

The matrix

$$\begin{bmatrix} \mathbf{W}^{-1} & \tilde{\mathbf{H}} \\ \tilde{\mathbf{H}}^T & \mathbf{0} \end{bmatrix}, \quad (45)$$

is always nonsingular [43]; hence, we have the optimal solution $(\tilde{\mathbf{z}}^*, \boldsymbol{\lambda}^*)$ given by

$$\begin{bmatrix} \tilde{\mathbf{z}}^* \\ \boldsymbol{\lambda}^* \end{bmatrix} = \begin{bmatrix} \mathbf{W}^{-1} & \tilde{\mathbf{H}} \\ \tilde{\mathbf{H}}^T & \mathbf{0} \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{0} \\ \tilde{\mathbf{e}} \end{bmatrix}. \quad (46)$$

By using the fact that

$$\begin{bmatrix} \mathbf{W}^{-1} & \tilde{\mathbf{H}} \\ \tilde{\mathbf{H}}^T & \mathbf{0} \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{0} & \tilde{\mathbf{H}}^{-T} \\ \tilde{\mathbf{H}}^{-1} & \mathbf{W}^{-1} \tilde{\mathbf{H}}^{-T} \end{bmatrix}, \quad (47)$$

the unique solutions to (46) are

$$\tilde{\mathbf{z}}^* = -\mathbf{W}(\tilde{\mathbf{H}}\boldsymbol{\lambda}^*), \quad (48)$$

$$\boldsymbol{\lambda}^* = -(\tilde{\mathbf{H}}^T \mathbf{W}^{-1} \tilde{\mathbf{H}})^{-1} \tilde{\mathbf{e}}. \quad (49)$$

Substituting (49) into (48) and computing $\tilde{\mathbf{z}}^{*T}$ allows us to form the product $\tilde{\mathbf{z}}^{*T} \mathbf{W}^{-1} \tilde{\mathbf{z}}^*$ as

$$\tilde{\mathbf{z}}^{*T} \mathbf{W}^{-1} \tilde{\mathbf{z}}^* = \tilde{\mathbf{e}}^T \tilde{\mathbf{H}}^{-1} \mathbf{W}^{-1} \tilde{\mathbf{H}}^{-T} \tilde{\mathbf{e}}$$

⁵ $\mathbf{R}^{-1}$ is positive definite so its square root always exists.

$$= \tilde{\mathbf{e}}^T \left(\tilde{\mathbf{H}}^T \mathbf{W} \tilde{\mathbf{H}} \right)^{-1} \tilde{\mathbf{e}}, \quad (50)$$

which is the objective of Problem 1 with $\tilde{\mathbf{Q}} = \tilde{\mathbf{H}}^T \text{diag}(\tilde{\boldsymbol{\mu}}) \tilde{\mathbf{H}} = \tilde{\mathbf{H}}^T \mathbf{W} \tilde{\mathbf{H}}$. Substituting $\tilde{\mathbf{z}}^*$ into the linear equality constraint $\tilde{\mathbf{H}}^T \tilde{\mathbf{z}} = \tilde{\mathbf{e}}$ simply verifies the identity $\tilde{\mathbf{e}} = \tilde{\mathbf{e}}$. Therefore, by solving Problem 2, isolating the variable $\tilde{\mathbf{z}}$, substituting it back to (42), and optimizing over $\tilde{\boldsymbol{\mu}}$ we arrive at Problem 1, which settles their equivalence (i.e. optimizing Problem 2 over $\tilde{\mathbf{z}}$ and $\tilde{\boldsymbol{\mu}}$ results in Problem 1). $\square$

Since we have established the equivalence of Problems 1 and 2, we now further manipulate Problem 2 into the final desired form. Introducing variables $s_i \geq 0$ we get

$$\begin{aligned} & \underset{\substack{s_1, s_2, \dots, s_r \\ \tilde{\mu}_1, \tilde{\mu}_2, \dots, \tilde{\mu}_r \\ \tilde{z}_1, \tilde{z}_2, \dots, \tilde{z}_r}}{\text{minimize}} & \sum_{i=1}^r s_i \\ & \text{subject to} & \frac{\tilde{z}_i^2}{2\tilde{\mu}_i} \leq s_i, \quad s_i \geq 0, \\ & & \tilde{\mathbf{H}}^T \tilde{\mathbf{z}} = \tilde{\mathbf{e}}, \\ & & \sum_{i=1}^r \tilde{\mu}_i = nk, \\ & & 0 \leq \tilde{\mu}_i \leq 1, \quad i = 1, \dots, r. \end{aligned} \quad (51)$$

The constraint $\frac{\tilde{z}_i^2}{2\tilde{\mu}_i} \leq s_i$ defines a set of hyperbolic constraints [44] that can be written as

$$\left\| \begin{bmatrix} 2\tilde{z}_i \\ 2s_i - \tilde{\mu}_i \end{bmatrix} \right\|_2 \leq 2s_i + \tilde{\mu}_i, \quad s_i \geq 0, \quad i = 1, \dots, r, \quad (52)$$

which are a set of second-order cone constraints. We can now state the desired optimization problem as a second-order cone program as

$$\begin{aligned} & \underset{\substack{s_1, s_2, \dots, s_r \\ \tilde{\mu}_1, \tilde{\mu}_2, \dots, \tilde{\mu}_r \\ \tilde{z}_1, \tilde{z}_2, \dots, \tilde{z}_r}}{\text{minimize}} & \sum_{i=1}^r s_i \\ & \text{subject to} & \left\| \begin{bmatrix} 2\tilde{z}_i \\ 2s_i - \tilde{\mu}_i \end{bmatrix} \right\|_2 \leq 2s_i + \tilde{\mu}_i, \quad s_i \geq 0, \\ & & \tilde{\mathbf{H}}^T \tilde{\mathbf{z}} = \tilde{\mathbf{e}}, \\ & & \sum_{i=1}^r \tilde{\mu}_i = nk, \\ & & 0 \leq \tilde{\mu}_i \leq 1, \quad i = 1, \dots, r. \end{aligned} \quad (53)$$

As a final step, we convert back to our original set of variables to directly solve for the desired μ_i s. This step simply reshapes the relevant arrays and does not change the problem. For instance, $\tilde{\mathbf{z}} = \text{reshape}(\tilde{\mathbf{z}}, (n, m))$, $\mathbf{s} = \text{reshape}(s, (n, m))$. The required

form is

$$\begin{aligned} & \underset{\substack{s_{ij} \in \mathbb{R}_+ \\ \mu_j \in \mathbb{R}^m \\ \tilde{z}_{ij} \in \mathbb{R}}}{\text{minimize}} & \sum_{i=1}^n \sum_{j=1}^m s_{ij} \\ & \text{subject to} & \left\| \begin{bmatrix} 2\tilde{z}_{ij} \\ 2s_{ij} - \mu_j \end{bmatrix} \right\|_2 \leq 2s_{ij} + \mu_j, \quad s_{ij} \geq 0, \\ & & \tilde{\mathbf{H}}^T \mathbf{z} = \mathbf{e}_1, \\ & & \vdots \\ & & \tilde{\mathbf{H}}^T \mathbf{z} = \mathbf{e}_n, \\ & & \sum_{j=1}^m \mu_j = k, \\ & & 0 \leq \mu_j \leq 1, \quad j = 1, \dots, m. \end{aligned} \quad (54)$$

The solution to (54) yields an m vector $\boldsymbol{\mu}^*$ from which we take the sources with the k largest values as the desired selection. The solution can readily be obtained from most optimization solvers. The SOCP formulation (54) can be solved significantly faster than the SDP form in (35). This is due to the reduced complexity of SOCPs compared to SDPs [45]. Moreover, while solving SDPs can be computationally demanding for deployment on embedded systems, SOCPs can be solved on such platforms [46], [47]. We demonstrate this very aspect on an ARM-based embedded platform in Section IV. The comparative results of the SOCP formulation with the SDP and other state-of-the-art methods are discussed in Section IV-B.

3) Exact Solution With Mixed-Integer SOCP Formulation:

The relaxed SOCP problem in (54) can be put into a setting that solves the selection problem *exactly* by setting up the problem as a mixed-integer SOCP given by

$$\begin{aligned} & \underset{\substack{s_{ij} \in \mathbb{R}_+ \\ \mu_j \in \{0,1\} \\ \tilde{z}_{ij} \in \mathbb{R}}}{\text{minimize}} & \sum_{i=1}^n \sum_{j=1}^m s_{ij} \\ & \text{subject to} & \left\| \begin{bmatrix} 2\tilde{z}_{ij} \\ 2s_{ij} - \mu_j \end{bmatrix} \right\|_2 \leq 2s_{ij} + \mu_j, \quad s_{ij} \geq 0, \\ & & \tilde{\mathbf{H}}^T \mathbf{z} = \mathbf{e}_1, \\ & & \vdots \\ & & \tilde{\mathbf{H}}^T \mathbf{z} = \mathbf{e}_n, \\ & & \sum_{j=1}^m \mu_j = k, \\ & & \mu_j \in \{0,1\}, \quad j = 1, \dots, m. \end{aligned} \quad (55)$$

We note the constraint $\mu_j \in \{0,1\}$ that is set up to decide the inclusion or exclusion of a satellite from the selection. This problem is solvable using mixed-integer conic solvers such as the one in [48]. Such tools use an efficient branch and bound

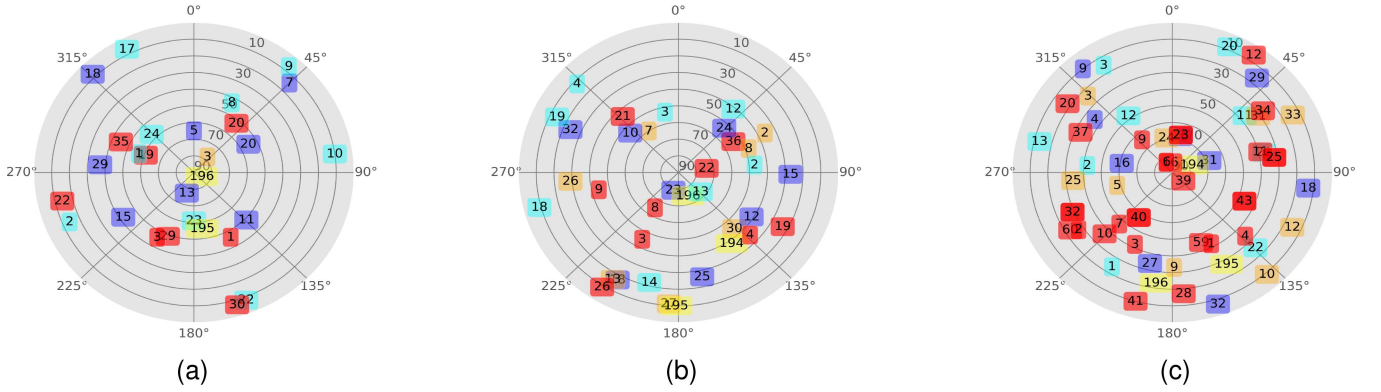


Fig. 4. Sky plot for GNSS satellites in view. GPS in blue, GLONASS in cyan, QZSS in yellow, Beidou in red, and Galileo in orange. The center point of the plot represents the user's location. Concentric circles moving outward from the user represent decreasing elevation of satellites. The azimuth is measured clockwise from 0°. (a) $m = 28$ satellites in view. (b) $m = 35$ satellites in view. (c) $m = 59$ satellites in view.

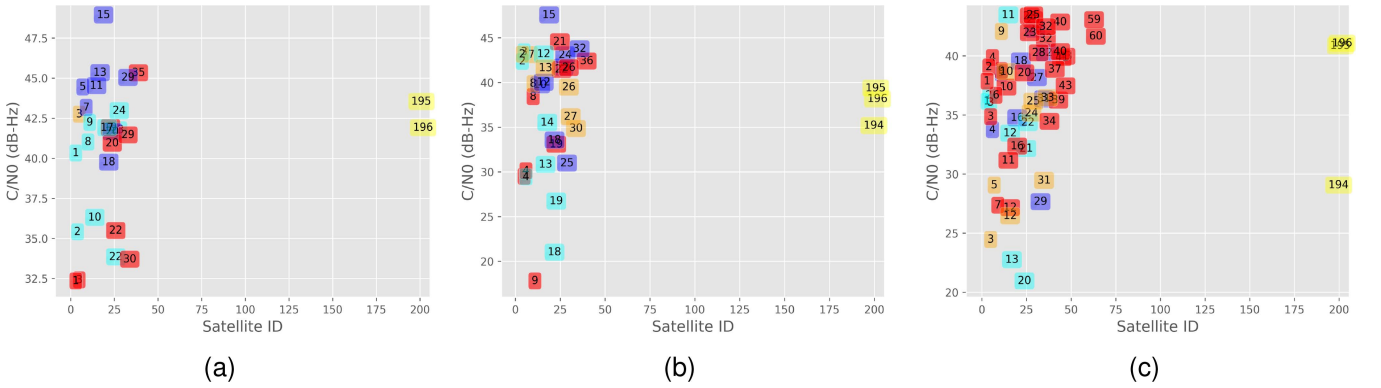


Fig. 5. C/N_0 distribution of satellites in view. GPS in blue, GLONASS in cyan, QZSS in yellow, Beidou in red, and Galileo in orange. (a) $m = 28$ satellites in view. (b) $m = 35$ satellites in view. (c) $m = 59$ satellites in view.

(B & B) method for a solution. The reduction to SOCP which we performed in Section III-D2 was the key step in enabling the exact solution to the problem.

IV. EXPERIMENTAL STUDIES

A. Setup

Real-world multi-constellation GNSS data for experimental validation was collected at 36.3755° latitude north, and 127.3607° longitude east, Daejeon, South Korea. The software tool discussed in [49] was utilized. To show the consistency of results across different data, three separate instances were taken. In the first case, a modest set (by current standards) of 28 satellites in view was considered. 35 satellites in view were considered in the second case. The third case contains a dense set of satellites in view, numbering 59 in total. The respective sky plots for the three cases are shown in Fig. 4, while the C/N_0 values of the satellite signals are shown in Fig. 5.

After extracting the relevant parameters from the data, the solutions to the SDP problem in (35), the SOCP problem in (54), and the mixed-integer SOCP (55) were obtained using the tool in [50] to interface with appropriate conic solvers. The proposed methods and competing approaches (greedy search

TABLE I
SPECIFICATIONS OF EMBEDDED PLATFORM

Platform	Processor architecture	RAM (GB) (usable)	Operating system
Raspberry Pi 4 (Model B)	Cortex-A72 (ARM v8) 1.8GHz	4	Linux

and genetic algorithm) are all compared using the WGDOP metric introduced in Section II-D as well as computation time. A smaller value of the WGDOP metric is the desired outcome. The experiments were run on a desktop computer running at 3.7 GHz. Furthermore, we also provide an implementation of the approximate SOCP selection in (54) on an embedded platform to demonstrate our method's usability on lower-power computational devices. The chosen platform is the Raspberry Pi 4 shown in Fig. 6. Its specifications are given in Table I.⁶

⁶Further detailed information can be found at <https://www.raspberrypi.com/products/raspberry-pi-4-model-b/specifications/>.

TABLE II
COMPARATIVE PERFORMANCE OF *A*-OPTIMAL SDP, SOCP, BRUTE-FORCE, GREEDY SELECTION, AND GENETIC ALGORITHM (GA) ($m = 28$)

Selection type (k -best)	SDP & SOCP (proposed)		Brute-force optimal		Greedy search		GA	All-in-view
	Selection	WGDOP	Selection	WGDOP	Selection	WGDOP	WGDOP	WGDOP
6	29-G, 17-GLO, 30-B, 196-Q, 7-G, 22-GLO	15.9936	29-G, 17-GLO, 30-B, 196-Q, 7-G, 22-GLO	15.9936	29-G, 17-GLO, 30-B, 196-Q, 7-G, 22-GLO	15.9936	16.2678	8.5853
8	9-GLO, 2-GLO, 29-G, 17-GLO, 7-G, 196-Q, 30-B, 22-GLO	13.8965	9-GLO, 2-GLO, 29-G, 17-GLO, 7-G, 196-Q, 30-B, 22-GLO	13.8965	9-GLO, 2-GLO, 29-G, 17-GLO, 7-G, 196-Q, 30-B, 22-GLO	13.8965	13.9649	
12	29-G, 11-G, 5-G, 22-B, 10-GLO, 196-Q, 9-GLO, 7-G, 30-B, 22-GLO, 17-GLO, 2-GLO	11.4174	1-B, 29-G, 3-GAL, 196-Q, 2-GLO, 18-G, 17-GLO, 30-B, 22-GLO, 9-GLO, 7-G, 10-GLO	11.3624	1-B, 29-G, 3-GAL, 196-Q, 2-GLO, 18-G, 17-GLO, 30-B, 22-GLO, 9-GLO, 7-G, 10-GLO	11.3624	11.4646	
24	29-B, 24-GLO, 23-GLO, 20-B, 35-B, 19-B, 15-G, 5-G, 1-B, 8-GLO, 29-G, 11-G, 3-GAL, 13-G, 196-Q, 22-B, 2-GLO, 18-G, 17-GLO, 7-G, 9-GLO, 30-B, 22-GLO, 10-GLO	8.8445	3-B, 23-GLO, 24-GLO, 20-B, 35-B, 19-B, 15-G, 1-B, 5-G, 11-G, 8-GLO, 29-G, 13-G, 3-GAL, 196-Q, 22-B, 2-GLO, 18-G, 17-GLO, 22-GLO, 30-B, 7-G, 9-GLO, 10-GLO	8.8438	29-B, 24-GLO, 23-GLO, 20-B, 35-B, 19-B, 15-G, 5-G, 1-B, 8-GLO, 29-G, 11-G, 3-GAL, 13-G, 196-Q, 22-B, 2-GLO, 18-G, 17-GLO, 7-G, 9-GLO, 30-B, 22-GLO, 10-GLO	8.8445	8.8438	

Key: G-GPS, GLO-GLONASS, Q-QZSS, B-Beidou, GAL-Galileo.

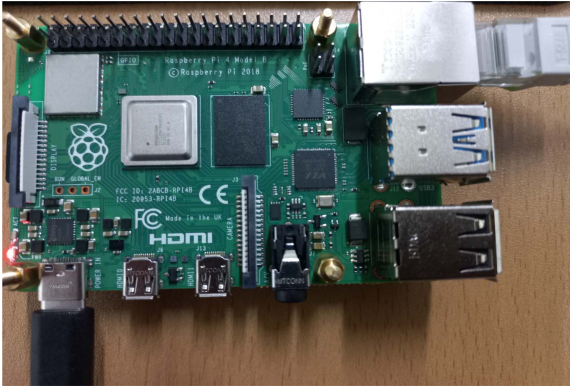


Fig. 6. Embedded platform for testing approximate SOCP-based GNSS satellite selection.

B. Results and Discussion

For the three cases considered, the results of the selections and their corresponding WGDOP values are shown in Tables II, III, and IV. The WGDOP values for the use of all-in-view satellites are also presented. For $m = 28$, a comparison with the absolute best solution using a brute-force search strategy is shown. For $m = 35$ and $m = 59$, except for the small values of subsets k , it is impractical to compute the brute-force solution. For instance, to select the best 12 satellites from $m = 35$, one would need to search through 834,451,800 possibilities and compare WGDOP values. The brute-force method is presented to provide the reader with some context of the challenge of the problem. For instance, for $m = 28$ and $k = 12$, it took roughly 10 days to solve the problem via brute force using standard hardware.

One of the proposed methods, mixed-integer SOCP, yields exact solutions to the selection problem; therefore, we use it for the case of $m = 35$ and $m = 59$. The plot of WGDOP against the number of satellites for all three experimental scenarios is shown in Fig. 7. From this figure, we can observe that, after

a certain number, adding more satellites has minimal effect on WGDOP, hence justifying the need for a carefully selected set of satellites rather than using an all-in-view solution. An efficient greedy search method (explained in the introduction) is also compared as a candidate.

The brute-force method considers all ${}^m C_k$ options and selects the globally optimal solution. We recall from the introduction section that the greedy search strategy works by removing one satellite at a time and evaluating the optimization objective using the resulting geometry [3]. For m satellites and k subsets, $N = 0.5(m(m+1) - k(k+1))$ objectives need to be evaluated. This method is known to give good results and is computationally efficient. For $m = 28$, the results show that the proposed SOCP method can attain the global optimum for $k = 6$ and $k = 8$ while being only 0.4817% and 0.0079% suboptimal for $k = 12$ and $k = 24$ respectively. The other proposed method, mixed-integer SOCP, gives exact solutions at the expense of marginally higher computational cost. A genetic algorithm (GA) is also considered for comparison. A weighted objective is designed by taking the inverse of WGDOP as a fitness function and using a population size of 8000 (for a reasonable compromise between speed and accuracy). Due to space constraints, we only show the WGDOP values for the GA, not the specific selections. Computation time for the proposed set of optimization-based methods, compared with that of GA and greedy search are given in Table V. The computation times for the GA-based approach are rather long (more than three orders of magnitude longer than the fastest proposed method). The proposed SOCP approach significantly outperforms all the considered methods. Moreover, we show in Table VI that the SOCP solution can be obtained within a short time even on a much less powerful computational platform. This greatly increases the utility of the proposed method in practical settings.

A comment on the advantages of the weighted objective pursued in this research is necessary. In Fig. 8, we show a comparative plot of multipath, ionospheric, tropospheric, and

TABLE III
COMPARATIVE PERFORMANCE OF *A*-OPTIMAL SDP, SOCP, MIXED-INTEGER SOCP, GREEDY SELECTION, AND GENETIC ALGORITHM (GA) ($m = 35$)

Selection type (k -best)	SDP & SOCP (proposed)		Mixed-integer SOCP (proposed, exact)		Greedy search		GA	All-in-view
	Selection	WGDOP	Selection	WGDOP	Selection	WGDOP	WGDOP	WGDOP
6	32-G, 27-GAL, 4-GLO, 15-G, 23-G, 195-Q	16.6482	2-GAL, 27-GAL, 195-Q, 32-G, 4-GLO, 23-G	16.2821	23-G, 12-GLO, 15-G, 32-G, 195-Q, 26-B	16.4024	16.4024	8.1711
8	196-Q, 12-GLO, 32-G, 4-GLO, 23-G, 27-GAL, 195-Q, 15-G	14.0914	196-Q, 12-GLO, 32-G, 4-GLO, 23-G, 27-GAL, 195-Q, 15-G	14.0914	23-G, 12-GLO, 15-G, 32-G, 19-B, 4-GLO, 195-Q, 26-B	14.2509	14.0914	
12	2-GAL, 19-GLO, 15-G, 19-B, 26-B, 23-G, 196-Q, 195-Q, 32-G, 27-GAL, 4-GLO, 12-GLO	11.6079	2-GAL, 19-GLO, 15-G, 19-B, 26-B, 23-G, 196-Q, 195-Q, 32-G, 27-GAL, 4-GLO, 12-GLO	11.6079	23-G, 196-Q, 8-B, 12-GLO, 15-G, 32-G, 19-B, 19-GLO, 4-GLO, 27-GAL, 194-Q, 26-B	11.6177	11.6079	
24	4-B, 7-GAL, 8-GAL, 15-G, 2-GAL, 19-B, 14-GLO, 19-B, 26-B, 32-G, 13-GLO, 13-GAL, 22-B, 18-G, 196-Q, 8-B, 23-G, 26-GAL, 27-GAL, 4-GLO, 195-Q, 25-G, 12-GLO, 21-B	8.8075	4-B, 7-GAL, 8-GAL, 15-G, 2-GAL, 19-B, 14-GLO, 19-B, 26-B, 32-G, 13-GLO, 13-GAL, 22-B, 18-G, 196-Q, 8-B, 23-G, 26-GAL, 27-GAL, 4-GLO, 195-Q, 25-G, 12-GLO, 21-B	8.8075	4-B, 7-GAL, 8-GAL, 15-G, 2-GAL, 19-B, 14-GLO, 19-B, 26-B, 32-G, 13-GLO, 13-GAL, 22-B, 18-G, 196-Q, 8-B, 23-G, 26-GAL, 27-GAL, 4-GLO, 195-Q, 25-G, 12-GLO, 21-B	8.8075	8.8075	

Key: G-GPS, GLO-GLONASS, Q-QZSS, B-Beidou, GAL-Galileo.

TABLE IV
COMPARATIVE PERFORMANCE OF *A*-OPTIMAL SDP, SOCP, MIXED-INTEGER SOCP, GREEDY SELECTION, AND GENETIC ALGORITHM (GA) ($m = 59$)

Selection type (k -best)	SDP & SOCP (proposed)		Mixed-integer SOCP (proposed, exact)		Greedy search		GA	All-in-view
	Selection	WGDOP	Selection	WGDOP	Selection	WGDOP	WGDOP	WGDOP
6	18-G, 3-GLO, 6-B, 33-GAL, 20-B, 196-Q	16.1258	12-B, 195-Q, 60-B, 33-GAL, 20-B, 6-B	15.8751	196-Q, 3-GLO, 6-B, 10-GAL, 20-B, 34-B	16.0325	15.9617	5.9295
8	39-B, 60-B, 18-G, 3-GLO, 196-Q, 20-B, 6-B, 33-GAL	13.7894	196-Q, 28-B, 37-B, 34-B, 33-GAL, 20-B, 6-B, 39-B	13.7379	196-Q, 3-GLO, 60-B, 39-B, 6-B, 10-GAL, 20-B, 34-B	13.9769	13.7781	
12	10-GAL, 28-B, 9-G, 60-B, 196-Q, 39-B, 6-B, 3-GLO, 20-B, 34-B, 33-GAL, 18-G	11.1519	32-B, 196-Q, 28-B, 32-G, 18-G, 34-B, 33-GAL, 20-B, 9-G, 3-GLO, 6-B, 39-B	11.1517	196-Q, 3-GLO, 60-B, 39-B, 12-B, 6-B, 10-GAL, 33-GAL, 20-B, 28-B, 34-B, 37-B	11.2250	11.2104	
24	2-B, 195-Q, 40-B, 25-B, 20-B, 37-B, 60-B, 9-G, 4-G, 3-GLO, 41-B, 31-G, 28-B, 196-Q, 32-G, 39-B, 6-B, 10-GAL, 16-B, 18-G, 12-B, 29-G, 33-GAL, 34-B	8.1010	2-B, 195-Q, 40-B, 25-B, 20-B, 37-B, 60-B, 9-G, 4-G, 3-GLO, 41-B, 31-G, 28-B, 196-Q, 32-G, 39-B, 6-B, 10-GAL, 16-B, 18-G, 12-B, 29-G, 33-GAL, 34-B	8.1010	2-B, 195-Q, 40-B, 25-B, 20-B, 37-B, 60-B, 9-G, 4-G, 3-GLO, 41-B, 31-G, 28-B, 196-Q, 32-G, 39-B, 6-B, 10-GAL, 16-B, 18-G, 12-B, 29-G, 33-GAL, 34-B	8.1010	8.1364	

Key: G-GPS, GLO-GLONASS, Q-QZSS, B-Beidou, GAL-Galileo.

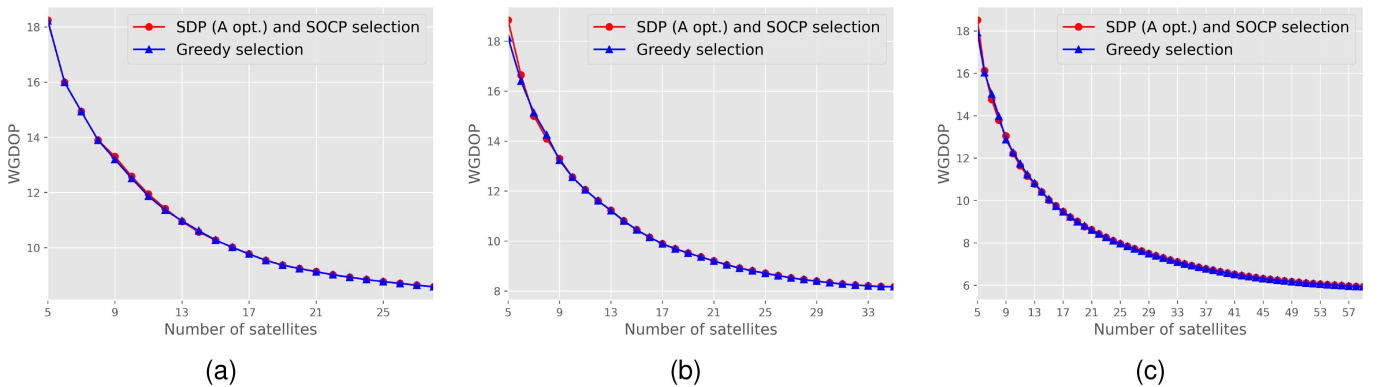


Fig. 7. WGDOP performance of *A*-optimal SDP, SOCP, and greedy search-based selection for a varying number of satellites (a) $m = 28$ satellites in view (b) $m = 35$ satellites in view (c) $m = 59$ satellites in view.

TABLE V
COMPUTATION TIMES (IN MILLISECONDS) FOR GREEDY SEARCH (GS), GENETIC ALGORITHM (GA), A-OPTIMAL SDP, THE PROPOSED SOCP AND MIXED-INTEGER SOCP (MI-SOCP) SOLUTIONS, k IS THE NUMBER OF SELECTED SATELLITES

k	$m = 28$					$m = 35$					$m = 59$				
	GS	GA	SDP	SOCP	MI-SOCP	GS	GA	SDP	SOCP	MI-SOCP	GS	GA	SDP	SOCP	MI-SOCP
6	24.00	4439.71	44.00	3.011	161.85	30.03	4795.01	46.00	5.017	205.71	152.00	18558.08	69.96	6.021	1283.07
8	18.03	4559.41	44.99	3.009	351.42	33.02	4931.88	49.86	6.021	128.43	104.99	18102.12	61.00	7.024	511.74
12	18.02	4564.18	43.99	3.007	228.40	32.67	4926.99	49.71	4.014	165.56	154.99	18097.11	64.96	5.017	593.76
24	6.00	5053.48	42.99	3.307	56.19	19.99	5913.03	45.99	5.016	73.25	96.02	20199.48	67.00	5.018	269.92

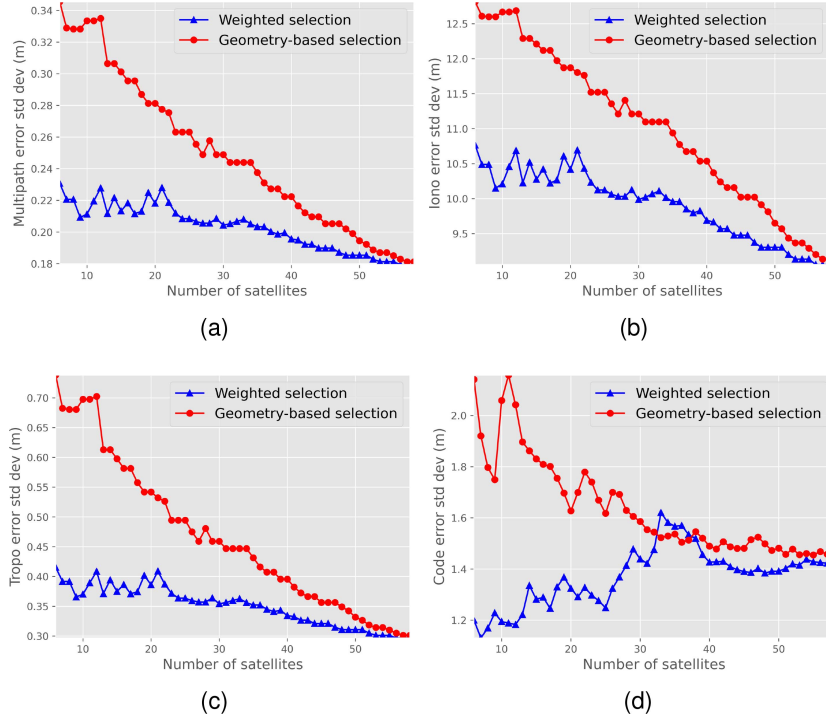


Fig. 8. Performance comparison of weighted satellite selection with that of a geometric selection procedure for $m = 59$. It can be observed that the weighted selection approach provides significant benefits in reducing, (a) multipath error, (b) ionospheric error, (c) tropospheric error, and (d) code error.

TABLE VI
COMPUTATION TIMES (IN MILLISECONDS) FOR A C PROGRAMMING LANGUAGE-BASED IMPLEMENTATION OF SOCP SATELLITE SELECTION ON A RASPBERRY PI 4 PLATFORM

Selection type (k -best)	$m = 28$	$m = 35$	$m = 59$
6	259.661	280.847	564.133
12	65.717	79.255	170.955
24	38.684	48.039	54.256

code errors from the proposed approach with that of purely geometry-based methods. We take the case of 59 satellites in view. From the plots, we can see that selections using the weighted metric perform significantly better than those of the geometric approach for each type of error. This is due to the fact the proposed approach optimizes overall performance instead of just geometric diversity. As a result, we get a mean of 21.67%, 10.85%, 30.99%, and 21.65%, reduction in multipath error,

ionospheric error, tropospheric error, and code error respectively.

C. Additional Formulations

1) *Penalty-Based Selection*: One benefit of optimization-based approaches in general is the ease of problem modifications to fit new specifications. In the current context, suppose we want to further increase the elevation of the satellites to be selected due to prior knowledge that the user is located in a deep urban canyon. This might require us to select satellites with elevations even higher than the ones selected with the WGDOP objective in this paper. We can extend the framework by considering a penalty matrix $C \in \mathbb{S}_{++}^n$ that further discourages the use of lower elevation satellites. We define a penalized WGDOP (PWGDOP) metric as

$$\text{PWGDOP} = \sqrt{\text{Tr}(\mathbf{H}^T \mathbf{C} \mathbf{R}^{-1} \mathbf{H})^{-1}}. \quad (56)$$

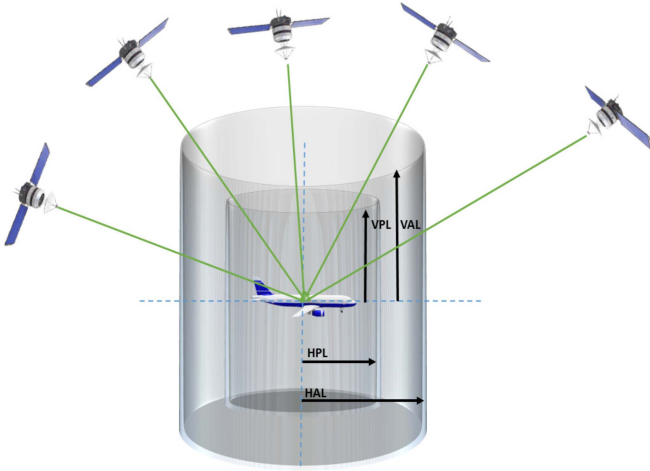


Fig. 9. Definitions of VPL, VAL, HPL, and HAL for aviation applications.

C is a user-selected parameter that depends on the application. For instance, if we set $C = R^{-1}$, for the case of $m = 28$ and $k = 6$, using either the SDP or SOCP approach, we get the selection $\{15\text{-G}, 7\text{-G}, 29\text{-G}, 11\text{-G}, 8\text{-GLO}, 196\text{-Q}\}$. This set is at a much higher elevation than the ones selected using WGDOP, potentially enabling better localization for urban scenarios.⁷

2) *Application to Safety-Critical Navigation Systems*: In applications, such as aviation, where safety is of high priority, it is essential to select navigation satellites to meet certain margins. The horizontal protection level (HPL) and vertical protection level (VPL) are quantities that define a region around the true user location which is guaranteed to contain the position solution. These quantities should be within their respective alert limits (horizontal alert limit (HAL) and vertical alert limit (VAL)). Fig. 9 illustrates the concept. In applications, $VPL > HPL$ generally, and meeting VPL requirements is more important than that of HPL. Using the components of the covariance P in (12), the expression for VPL is given by [3]

$$VPL = K_v \sqrt{P_{33}}, \quad (57)$$

with $K_v = 5.33$, whereas the expression for HPL is

$$HPL = K_h \sqrt{\frac{P_{11} + P_{22}}{2} + \sqrt{\left(\frac{P_{11} - P_{22}}{4}\right)^2 + P_{12}^2}}, \quad (58)$$

where K_h is taken to be 6.0. The metric in (29) is useful for this particular case as it provides a selection that minimizes the worst-case estimation error. With this metric as an objective, we solve the relaxed version of the optimization problem in (32). The resulting solution will be an E -optimal selection with a weighted objective. Its solution for the case of $m = 28$ gives the results in Fig. 10. In Fig. 10(c), we note the distinctive advantage of the E -optimal selection in reducing VPL. This approach minimizes worst-case error variance in the min-max sense, giving lower VPL values. However, the reduction in VPL

⁷This advantage comes at the expense of higher horizontal dilution of precision (HDOP).

comes at the expense of HPL performance. Nonetheless, due to the bigger margins that a designer has for HPL, this compromise is acceptable and results in a selection that gives safer margins overall.

V. CONCLUSION

A set of optimization-based solutions to the problem of satellite selection for multi-constellation GNSS have been presented. An attempt has been made to give a comprehensive account of the satellite selection problem. In addition to the typical geometry-based GDOP metric, the pseudo-range measurement precision of each satellite has been considered. Satellite elevation-dependent errors such as ionospheric propagation delay, tropospheric propagation delay, multipath, and code tracking errors (that arise due to receiver noise or external interference) have been included in the presented model. The combination of geometry and pseudo-range measurement precision culminated in the formulation of a Weighted GDOP (WGDOP) optimization objective. Minimizing a version of this objective subject to cardinality constraints framed satellite selection as a Boolean optimization problem. This problem was relaxed to a semidefinite programming (SDP) problem and then converted to a second-order cone programming (SOCP) problem. The results showed that the SOCP method provides the same quality of selections as the SDP method while being an order of magnitude faster. The method also outperforms a fast heuristic called greedy selection method and the state-of-the-art genetic algorithm approaches. Moreover, we have validated the solution of the approximate SOCP formulation on an embedded platform. The SOCP approach was further extended to a mixed-integer SOCP formulation which enabled the exact solution to the selection problem. This is a new result for this field of research and gives a definitive solution to the source selection problem at large. Additional context-specific modifications to the basic selection problem have been presented to demonstrate the flexibility of the presented method.

APPENDIX A

SDP FORM OF THE WEIGHTED A -OPTIMAL SELECTION PROBLEM

We restate the relaxed version of the weighted A -optimal selection problem given in (34).

$$\begin{aligned} & \underset{\mu_1, \mu_2, \dots, \mu_m}{\text{minimize}} \quad \text{Tr} \left(\sum_{i=1}^m \mu_i \frac{1}{\sigma_{\epsilon^i}^2} \mathbf{h}_i \mathbf{h}_i^T \right)^{-1} \\ & \text{subject to} \quad \sum_{i=1}^m \mu_i = k, \\ & \quad \quad \quad 0 \leq \mu_i \leq 1, \quad i = 1, \dots, m. \end{aligned} \quad (59)$$

For the trace operator, the following identity holds,

$$\text{Tr} \left(\sum_{i=1}^m \mu_i \frac{1}{\sigma_{\epsilon^i}^2} \mathbf{h}_i \mathbf{h}_i^T \right)^{-1} = \sum_{j=1}^n \mathbf{e}_j^T \left(\sum_{i=1}^m \mu_i \frac{1}{\sigma_{\epsilon^i}^2} \mathbf{h}_i \mathbf{h}_i^T \right)^{-1} \mathbf{e}_j, \quad (60)$$

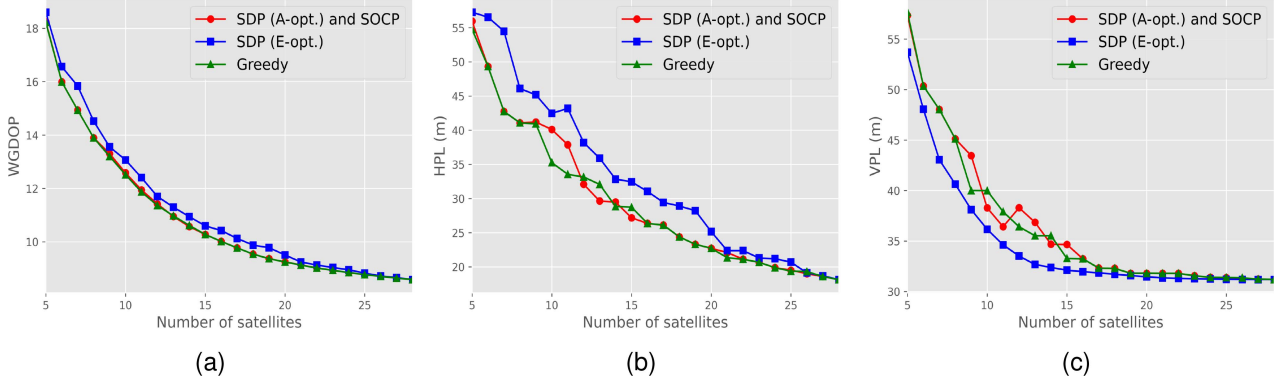


Fig. 10. Performance comparison of E -optimal, A -optimal, SOCP, and greedy search selection for $m = 28$ (a) WGDOP performance (b) HPL performance (c) VPL performance.

where e_j is the j -th unit vector. e_j selects columns while e_j^T selects rows of the matrix $(\sum_{i=1}^m \mu_i \frac{1}{\sigma_{\bar{e}^i}^2} \mathbf{h}_i \mathbf{h}_i^T)^{-1}$. The form in (59) can be stated in epigraph form as

$$\begin{aligned} & \underset{\substack{\mu_1, \mu_2, \dots, \mu_m \\ u_1, u_2, \dots, u_n}}{\text{minimize}} && \sum_{j=1}^n u_j \\ & \text{subject to} && \sum_{j=1}^n e_j^T \left(\sum_{i=1}^m \mu_i \frac{1}{\sigma_{\bar{e}^i}^2} \mathbf{h}_i \mathbf{h}_i^T \right)^{-1} e_j - \sum_{j=1}^n u_j \preceq 0, \\ & && \sum_{i=1}^m \mu_i = k, \\ & && 0 \leq \mu_i \leq 1, \quad i = 1, \dots, m. \end{aligned} \quad (61)$$

To progress to the next steps we state the Schur complement lemma.

Definition 1: Let M be a symmetric block matrix such that

$$M = \begin{bmatrix} A & B \\ B^T & C \end{bmatrix}, \quad (62)$$

where $A \in \mathbb{S}^n$, $C \in \mathbb{S}^m$, and $B \in \mathbb{R}^{n \times m}$. The Schur complement matrix, S , of A in M is given by

$$S = C - B^T A^{-1} B \quad (63)$$

Lemma 2: Given (62), $M \succeq 0$ (respectively $M \succ 0$) if and only if $S \succeq 0$ (respectively $S \succ 0$).

Proof: Refer to [44, Ch. 4] $\square$

Following the results of Lemma 2, multiplying out by negative, and reversing signs in the first constraint in (61), we can observe that the relation

$$\sum_{j=1}^n u_j - \sum_{j=1}^n e_j^T \left(\sum_{i=1}^m \mu_i \frac{1}{\sigma_{\bar{e}^i}^2} \mathbf{h}_i \mathbf{h}_i^T \right)^{-1} e_j \succeq 0, \quad (64)$$

transforms to

$$\begin{bmatrix} \sum_{i=1}^m \mu_i \frac{1}{\sigma_{\bar{e}^i}^2} \mathbf{h}_i \mathbf{h}_i^T & e_j \\ e_j^T & u_j \end{bmatrix} \succeq 0, \quad j = 1, \dots, n. \quad (65)$$

Going from (65) to (64) requires that we form the Schur complement of the left side of (65) and add the resulting n equations. Substituting (65) into (61) we have the final desired SDP form as

$$\begin{aligned} & \underset{\substack{\mu_1, \mu_2, \dots, \mu_m \\ u_1, u_2, \dots, u_n}}{\text{minimize}} && \sum_{j=1}^n u_j \\ & \text{subject to} && \begin{bmatrix} \sum_{i=1}^m \mu_i \frac{1}{\sigma_{\bar{e}^i}^2} \mathbf{h}_i \mathbf{h}_i^T & e_j \\ e_j^T & u_j \end{bmatrix} \succeq 0, \quad j = 1, \dots, n. \\ & && \sum_{i=1}^m \mu_i = k, \\ & && 0 \leq \mu_i \leq 1, \quad i = 1, \dots, m, \end{aligned} \quad (66)$$

which is the result stated in (35).

APPENDIX B

COMMENTS ON COMPUTATIONAL COMPLEXITY

The complexity of the SDP method for the general selection method proposed in [28] is approximately $\mathcal{O}(m^3)$ with m being the number of sources. In practice, modern solvers use further optimization such that the actual complexity can be lowered. Assuming that primal-dual interior point methods are used for a solution. We can compute the approximate complexity for the SOCP-based solution. The approximate worst-case arithmetic complexity for the general second-order cone programming problem (25) is given by [51]

$$\text{Comp} \approx \mathcal{O}((p+1)^{1/2} n \left(n^2 + p + \sum_{i=0}^p n_i^2 \right)). \quad (67)$$

The problem in (53) can be transformed into standard form given in (25) as

$$\begin{aligned} & \underset{\substack{s_1, s_2, \dots, s_r \\ \mu_1, \mu_2, \dots, \mu_r \\ \bar{z}_1, \bar{z}_2, \dots, \bar{z}_r}}{\text{minimize}} && \sum_{i=1}^r s_i \end{aligned}$$

$$\begin{aligned}
& \text{subject to } \left\| \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & -\frac{1}{\sqrt{2}} \\ 0 & 2 & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \tilde{z}_i \\ s_i \\ \tilde{\mu}_i \end{bmatrix} \right\|_2 \leq (0, 2, 1)^T \begin{bmatrix} \tilde{z}_i \\ s_i \\ \tilde{\mu}_i \end{bmatrix} \\
& s_i \geq 0, \\
& \tilde{\mathbf{H}}^T \tilde{\mathbf{z}} = \tilde{\mathbf{e}}, \\
& \sum_{i=1}^r \tilde{\mu}_i = nk, \\
& 0 \leq \tilde{\mu}_i \leq 1, \quad i = 1, \dots, r. \quad (68)
\end{aligned}$$

Based on this, we can give an account of the complexity of the SOCP formulation. We set $r = mn$ as defined before, and neglect the term with the sum since the coefficient matrix is low dimensional. We then have

$$Comp \approx \mathcal{O}((mn + 1)^{1/2} mn((mn)^2 + mn)). \quad (69)$$

However, for our application n is small, and the m^3 term in (69) dominates. We finally have the approximation

$$Comp \approx \mathcal{O}(m^3), \quad (70)$$

This might at first glance seem to be similar to the complexity of the SDP method; however, this is the worst-case complexity. When primal-dual interior point methods are applied, the biggest computational effort lies in solving the Karush-Kuhn-Tucker (KKT) system of the form

$$\begin{bmatrix} \mathbf{B}^{-1} & \mathbf{F}^T \\ \mathbf{F} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{v} \\ \mathbf{w} \end{bmatrix} = - \begin{bmatrix} \mathbf{g} \\ \mathbf{h} \end{bmatrix}, \quad (71)$$

where $\mathbf{F}$ is the coefficient matrix of the equality constraints, the vector $(\mathbf{v}, \mathbf{w})$ is the Newton search direction, and the matrix $\mathbf{B}$ is related to the second-order cone constraints and has diagonal structure. For SOCPs, the matrix $\mathbf{B}$ has a special structure. In particular, it has a diagonal plus low-rank structure [52]. It has the form $\mathbf{D} + \mathbf{UV}$ such that

$$\mathbf{B} = \begin{bmatrix} \times & & & \\ & \times & & \\ & & \ddots & \\ & & & \times \end{bmatrix} + \begin{bmatrix} | & | & & \\ \times & \times & \dots & \\ | & | & & \end{bmatrix} \begin{bmatrix} \text{---} & \times & \text{---} \\ \text{---} & \times & \text{---} \\ \vdots & & \end{bmatrix}, \quad (72)$$

where $\mathbf{D}$ is a diagonal matrix, and the product $\mathbf{UV}$ is a low-rank matrix. If we also have a sparse matrix $\mathbf{F}$, the use of matrix inversion lemma gives significant computational savings that lead to solutions in much less than $\mathcal{O}(m^3)$ [40, Appendix C]. This is the key to the computational advantage of solving SOCPs over SDPs. Further computational issues in semidefinite programming are discussed in [53].

ACKNOWLEDGMENT

The authors would like to express their sincere gratitude to the anonymous reviewers of the initial manuscript. The discussion in the current version has greatly benefited from their comments.

REFERENCES

- [1] P. J. Teunissen and O. Montenbruck, Eds. *Springer Handbook of Global Navigation Satellite Systems*. New York, NY, USA: Springer, 2017.
- [2] Y. T. J. Morton, F. Van Diggelen, J. J. Spilker, and B. W. Parkinson, Eds. *Position, Navigation, and Timing Technologies in the 21st Century*, New Jersey, USA: Wiley, 2021.
- [3] T. Walter, J. Blanch, and V. Kropp, "Satellite selection for aviation users of multi-constellation SBAS," in *Proc. 29th Int. Tech. Meeting Satell. Division Inst. Navigation*, Portland, Oregon, USA, 2016, pp. 1350–1359.
- [4] F. Meng, S. Wang, and B. Zhu, "GNSS reliability and positioning accuracy enhancement based on fast satellite selection algorithm and RAIM in multiconstellation," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 30, no. 10, pp. 14–27, Oct. 2015.
- [5] D. A. Divis, "Handling an embarrassment of riches—too many satellites," Accessed: Nov. 19, 2024. [Online] Available: <https://insidengnss.com/handling-an-embarrassment-of-riches-too-many-satellites/>
- [6] P. F. Swaszek, R. J. Hartnett, K. C. Seals, and R. Swaszek, "A temporal algorithm for satellite subset selection in multi-constellation GNSS," in *Proc. 2017 Int. Tech. Meeting Inst. Navigation*, 2017, pp. 1147–1159.
- [7] J.-C. Juang, "Performance-based GNSS satellite selection: A linear matrix inequality (LMI) approach," *NAVIGATION: J. Inst. Navigation*, vol. 70, no. 2, Jun. 2023.
- [8] P. D. Groves, *Principles of GNSS Inertial and Multisensor Integrated Navigation Systems*, 2nd Ed., Norwood, MA, USA: Artech House, 2013.
- [9] T. Walter, J. Blanch, and P. Enge, "Reduced subset analysis for multi-constellation ARAIM," in *Proc. Int. Tech. Meeting Inst. Navigation*, 2014, pp. 89–98.
- [10] M. Joerger and B. Pervan, "Fault detection and exclusion using solution separation and chi-squared ARAIM," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 52, no. 2, pp. 726–742, Apr. 2016.
- [11] Y. Ge, Z. Wang, and Y. Zhu, "Reduced ARAIM monitoring subset method based on satellites in different orbital planes," *GPS Solutions*, vol. 21, pp. 1443–1456, Aug. 2017.
- [12] J. Blanch, T. Walter, and P. Enge, "Fixed subset selection to reduce advanced RAIM complexity," in *Proc. Int. Tech. Meeting Inst. Navigation*, 2018, pp. 88–98.
- [13] S. Luo, L. Wang, R. Tu, W. Zhang, J. Wei, and C. Chen, "Satellite selection methods for multi-constellation advanced RAIM," *Adv. Space Res.*, vol. 65, no. 5, pp. 1503–1517, Dec. 2020.
- [14] D. Gerbeth, M. Felux, M. S. Cîrciu, and M. Caamano, "Optimized selection of satellite subsets for a multi-constellation GBAS," in *Proc. Int. Tech. Meeting Inst. Navigation*, 2016, pp. 360–367.
- [15] F. S. Prol et al., "Position, navigation, and timing (PNT) through low earth orbit (LEO) satellites: A survey on current status, challenges, and opportunities," *IEEE Access*, vol. 10, pp. 83971–84002, 2022.
- [16] J. Guo, Y. Wang, X. Xie, and C. Sun, "A fast satellite selection algorithm for positioning in LEO constellation," *Adv. Space Res.*, vol. 73, no. 1, pp. 271–285, Oct. 2024.
- [17] D. Y. Hsu, "Relations between dilutions of precision and volume of the tetrahedron formed by four satellites," in *Proc. 1994 IEEE Position. Location Navigation Symp.*, 1994, pp. 669–676.
- [18] Z. Zuo-ya, H. Cheng, F. Chu-gang, and Z. Fei-peng, "Selection of GPS satellites for the optimum geometry," *Chin. Astron. Astrophys.*, vol. 28, no. 1, pp. 80–87, Mar. 2004.
- [19] D. Simon and H. El-Sherief, "Navigation satellite selection using neural networks," *Neurocomputing*, vol. 7, no. 3, pp. 247–258, Apr. 1995.
- [20] M. S. Phatak, "Recursive method for optimum GPS satellite selection," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 37, no. 2, pp. 751–754, Apr. 2001.
- [21] J. Song, G. Xue, and Y. Kang, "A novel method for optimum global positioning system satellite selection based on a modified genetic algorithm," *PLOS One*, vol. 11, no. 3, Mar. 2016, Art. no. e0150005.
- [22] N. Blanco-Delgado and F. D. Nunes, "Satellite selection method for multi-constellation GNSS using convex geometry," *IEEE Trans. Veh. Technol.*, vol. 59, no. 9, pp. 4289–4297, Nov. 2010.
- [23] T. Soiminen, P. Syrjärinne, S. Ali-Löytty, and C. Schmid, "Data-driven approach to satellite selection in multi-constellation GNSS receivers," in *Proc. IEEE 8th Int. Conf. Localization GNSS*, 2018, pp. 1–6.
- [24] E. Wang, C. Sun, J. Guo, P. Qu, T. Pang, and J. Zhang, "GNSS receiver satellite selection algorithm based on particle swarm optimization," in *Proc. IEEE 15th Conf. Ind. Electron. Appl.*, 2020, pp. 985–989.
- [25] Q. Yu, Y. Wang, and Y. Shen, "A fast GNSS satellite selection algorithm for continuous real-time positioning," *GPS Solutions*, vol. 26, no. 3, pp. 1–13, Apr. 2022.

- [26] C. Savas, "Unsupervised clustering of GNSS receiver measurements for satellite selection," in *Proc. IEEE 10th Int. Conf. Recent Adv. Air Space Technol.*, 2022, pp. 1–6.
- [27] N. S. Zewge, T. Kim, and H. Bang, "A convex optimization approach to satellite selection for global navigation satellite system (GNSS) receivers," in *Proc. IEEE 95th Veh. Technol. Conf.*, 2022, pp. 1–5.
- [28] S. Joshi and S. Boyd, "Sensor selection via convex optimization," *IEEE Trans. Signal Process.*, vol. 57, no. 2, pp. 451–462, Feb. 2009.
- [29] X. Shen and P. K. Varshney, "Sensor selection based on generalized information gain for target tracking in large sensor networks," *IEEE Trans. Signal Process.*, vol. 62, no. 2, pp. 363–375, Jan. 2014.
- [30] R. Herzog, I. Riedel, and D. Uciński, "Optimal sensor placement for joint parameter and state estimation problems in large-scale dynamical systems with applications to thermo-mechanics," *Optim. Eng.*, vol. 19, no. 3, pp. 591–627, Jun. 2018.
- [31] O. Talcoth, G. Risting, and T. Rylander, "Convex optimization of measurement allocation for magnetic tracking systems," *Optim. Eng.*, vol. 18, pp. 849–871, Dec. 2016.
- [32] F. Pukelsheim, "On linear regression designs which maximize information," *J. Stat. Plan. Inference*, vol. 4, no. 4, pp. 339–364, Jun. 1980.
- [33] V. V. Fedorov, *Theory of Optimal Experiments*, New York, NY, USA: Academic Press, 1972.
- [34] P. Misra and P. Enge, *Global Positioning System: Signals, Measurements, and Performance, Revised 2nd Edition*, Lincoln, MA, USA: Ganga-Jamuna Press, 2012.
- [35] J. Farrell, *Aided Navigation: GPS With High Rate Sensors*, New York, NY, USA: McGraw-Hill Inc., 2008.
- [36] T. Kailath, A. H. Sayed, and B. Hassibi, *Linear Estimation*, Upper Saddle River, NJ, USA: Prentice Hall, 2000.
- [37] C. Hegarty and E. Kaplan, *Understanding GPS/GNSS Principles and Applications*, 3rd ed., Norwood, MA, USA: Artech House, 2017.
- [38] N. Blanco-Delgado and F. D. Nunes, "Satellite selection based on WDOP concept and convex geometry," in *Proc. IEEE 5th ESA Workshop Satell. Navigation Technol. Eur. Workshop GNSS Signals Signal Process.*, 2010, pp. 1–8.
- [39] N. I. Ziedan, "Multipath and NLOS signals identification and satellite selection algorithms for multi-constellation receivers," in *Proc. 29th Int. Tech. Meeting Satell. Division Inst. Navigation*, 2016, pp. 521–533.
- [40] S. Boyd and L. Vandenberghe, *Convex Optimization*, Cambridge, U.K.: Cambridge Univ. Press, 2004.
- [41] R. Harman and T. Jurik, "Computing c-optimal experimental designs using the simplex method of linear programming," *Comput. Statist. Data Anal.*, vol. 53, no. 2, pp. 247–254, Jul. 2008.
- [42] G. Sagnol, "Optimal design of experiments with application to the inference of traffic matrices in large networks: Second-order cone programming and submodularity. optimization and control," Ph.D. dissertation, École Nationale Supérieure des Mines de Paris, Paris, France, 2010.
- [43] D. G. Luenberger and Y. Ye, *Linear and Nonlinear Programming*, 5th Ed., Switzerland: Springer, 2021.
- [44] G. C. Calafiore and L. El Ghaoui, *Optimization Models*, Cambridge, U.K.: Cambridge Univ. Press, 2014.
- [45] M. S. Lobo, L. Vandenberghe, S. Boyd, and H. Lebret, "Applications of second-order cone programming," *Linear Algebra Appl.*, vol. 284, no. 1, pp. 193–228, Mar. 1998.
- [46] A. Domahidi, E. Chu, and S. Boyd, "ECOS: An SOCP solver for embedded systems," in *Proc. IEEE 2013 Eur. Control Conf.*, 2013, pp. 3071–3076.
- [47] M. Schaller, G. Banjac, S. Diamond, A. Agrawal, B. Stellato, and S. Boyd, "Embedded code generation with CVXPY," *IEEE Contr. Syst. Lett.*, vol. 6, pp. 2653–2658, 2022.
- [48] MOSEK ApS, "MOSEK optimization suite, Version 10," 2023. [Online]. Available: <https://www.mosek.com/downloads/>
- [49] G. Fu, M. Khider, and F. van Diggelen, "Android raw GNSS measurement datasets for precise positioning," in *Proc. 33rd Int. Tech. Meeting Satell. Division Inst. Navigation*, 2020, pp. 1925–1937.
- [50] S. Diamond and S. Boyd, "CVXPY: A python-embedded modeling language for convex optimization," *J. Mach. Learn. Res.*, vol. 17, no. 1, pp. 2909–2913, Jan. 2016.
- [51] A. Ben-Tal and A. Nemirovski, *Lectures on Modern Convex Optimization: Analysis, Algorithms, and Engineering Applications*, Philadelphia, PA, USA: SIAM, 2001.
- [52] F. Alizadeh and D. Goldfarb, "Second-order cone programming," *Math. Program.*, vol. 95, no. 1, pp. 3–51, Jan. 2003.
- [53] L. Vandenberghe and S. Boyd, "Semidefinite programming," *SIAM Rev.*, vol. 38, no. 1, pp. 49–95, Dec. 1996.



air, land, sea, and space applications. He particularly studies how optimization methods can be utilized in enhancing navigation system performance.



attitude control and mission design. From 1999 to 2000, he was an Assistant Professor with Chungnam National University, Daejeon. Since 2001, he has been a Professor with the Korea Advanced Institute of Science and Technology, Daejeon. His research interests include attitude control, mission design, real-time control, guidance of spacecraft, and UAVs for various applications.

Natnael S. Zewge (Graduate Student Member, IEEE) received the B.Sc. degree in electrical and computer engineering from the Addis Ababa Institute of Technology, Addis Ababa, Ethiopia, the M.Sc. degree in electrical engineering and the Ph.D. degree in aerospace engineering from the Korea Advanced Institute of Science and Technology, Daejeon, South Korea. Since 2024, he has been a Postdoctoral Researcher with KAIST. His research interests include statistical signal processing and multisensor fusion techniques for autonomous navigation systems for

Hyochoong Bang received the B.S. and M.S. degrees in aeronautical engineering from Seoul National University, Seoul, South Korea, in 1985 and 1987, respectively, and the Ph.D. degree from Texas A&M University, College Station, TX, USA, in 1992. From 1992 to 1994, he was a Research Assistant Professor with U.S. Naval Postgraduate School, Monterey, CA, USA, and conducted research on spacecraft attitude control. From 1995 to 1999, he was with Korea Aerospace Research Institute, Daejeon, South Korea, and worked in the area of geosynchronous spacecraft attitude control and mission design. From 1999 to 2000, he was an Assistant Professor with Chungnam National University, Daejeon. Since 2001, he has been a Professor with the Korea Advanced Institute of Science and Technology, Daejeon. His research interests include attitude control, mission design, real-time control, guidance of spacecraft, and UAVs for various applications.